

Session 07

Introduction to Information Systems



FACULTY OF INFORMATION TECHNOLOGY
UNIVERSITY OF SCIENCE - HO CHI MINH CITY

Learning Objective

- ☐ Knowing the potential impact of information systems (IS) in business
- ☐ Understanding various aspects if IS:
 - ☐ Describe the components of computer-based information systems.
 - ☐ Describe the various types of information systems.
 - ☐ Identify the major information systems that support each organizational level.
 - ☐ Describe the fundamental of database technology
 - ☐ Describe how information resources are managed and discuss the roles of the information systems department and the end users.
- ☐ Knowing the training program of Information Systems Major

- ☐ Why learn about Information Systems
- ☐ What is an Information System
- ☐ Business Information Systems
- ☐ Database Fundamentals
- ☐ Careers in Information Systems
- ☐ Information Systems Major

Why Learn About Information System?

Why Learn About IS?

- There are many examples of the ways in which information systems and technologies are embedded in your lives, examples:
 - Register for classes
 - Take classes, and not just classes from your university
 - Access class syllabi, information, PowerPoints, and lectures
 - Research class papers and presentations
 - Conduct banking
 - Pay your bills
 - Research, shop, and buy products from companies or other people
 - Sell your “stuff ”
 - Search for, and apply for, jobs
 - Make your travel reservations (hotel, airline, rental car)

Why Learn About IS?

- Information systems are used in almost every profession, including:
 - Entrepreneurs
 - Small, medium to large multinationals
 - Production, finance, accounting, marketing, management

- Information systems are indispensable to help you to achieve your career goals

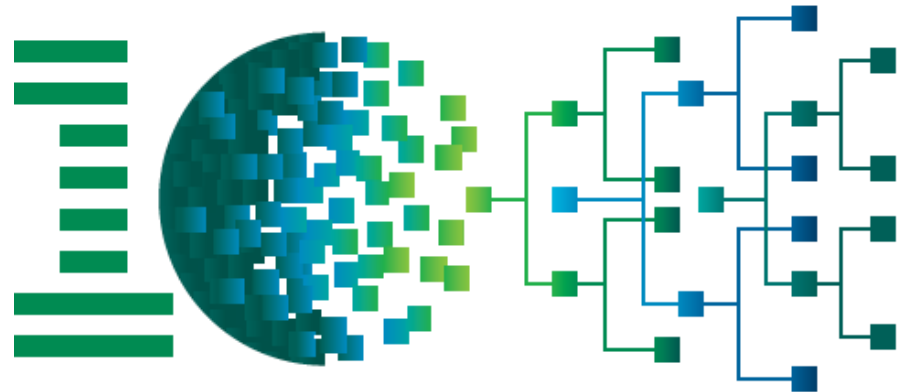
Why Learn About IS?

Position	Job Description
Chief Information Officer	Highest-ranking IS manager; is responsible for all strategic planning in the organization
IS Director	Manages all systems throughout the organization and the day-to-day operations of the entire IS organization
Information Center Manager	Manages IS services such as help desks, hot lines, training, and consulting
Applications Development Manager	Coordinates and manages new systems development projects
Project Manager	Manages a particular new systems development project
Systems Manager	Manages a particular existing system
Operations Manager	Supervises the day-to-day operations of the data and/or computer center
Programming Manager	Coordinates all applications programming efforts
Systems Analyst	Interfaces between users and programmers; determines information requirements and technical specifications for new applications
Business Analyst	Focuses on designing solutions for business problems; interfaces closely with users to demonstrate how IT can be used innovatively
Systems Programmer	Creates the computer code for developing new systems software or maintaining existing systems software
Applications Programmer	Creates the computer code for developing new applications or maintaining existing applications
Emerging Technologies Manager	Forecasts technology trends; evaluates and experiments with new technologies
Network Manager	Coordinates and manages the organization's voice and data networks
Database Administrator	Manages the organization's databases and oversees the use of database-management software
Auditing or Computer Security Manager	Oversees the ethical and legal use of information systems
Webmaster	Manages the organization's World Wide Web site
Web Designer	Creates World Wide Web sites and pages

What is an Information System?

What is an information system?

A set of interrelated *components* that
collect/retrieve, process, store and distribute
information
to *support decision making and control* in an
organization

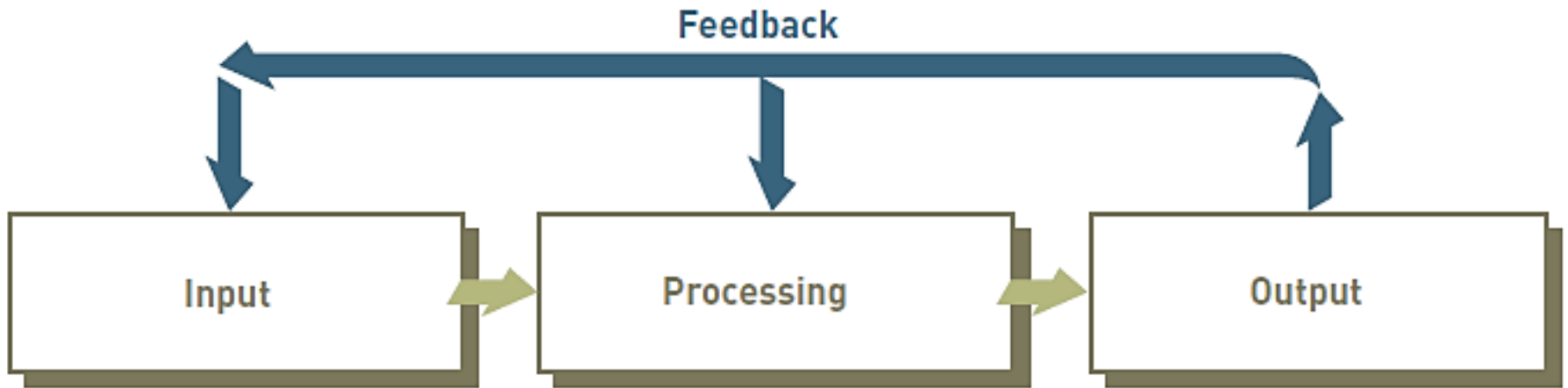


From ibmims

What is an Information System?

- An IS is a set of interrelated elements that:
 - Collect (input)
 - Manipulate (process)
 - Store
 - Disseminate (output) data and information
 - Provide a corrective reaction (feedback mechanism) to meet an objective

Components of an IS



Input, Processing, Storing, Output, Feedback

- Input: activity of gathering and capturing raw data
 - ▣ By person using a keyboard, scanner, mouse...
 - ▣ By computer, device, sensor (video, images, ...)
- Processing: converting data into useful outputs
 - ▣ i.e. purchase order → inventory reports
- Storing: computer data storage
 - ▣ Storing data for calculation
- Output: production of useful information
 - ▣ i.e. reports, documents, charts, graphs, pictures, etc.
- Feedback: information from the system
 - ▣ Used to make changes to input or processing activities

Concepts of Information System

- ☐ Data, Information and Knowledge
- ☐ Business process
- ☐ Information system development process
- ☐ Information system development tools

Data, Information, and Knowledge

- **Data:** streams of raw facts
- **Information:** collection of facts organized in such a way that they have value beyond the facts themselves. It is data shaped into meaning form
- **Knowledge:** awareness and understanding of someone or something which is acquired through experience or education by perceiving, discovering, or learning



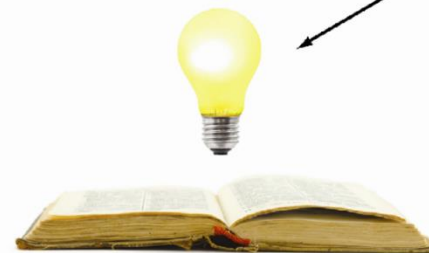
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Data



Exactostock/SuperStock

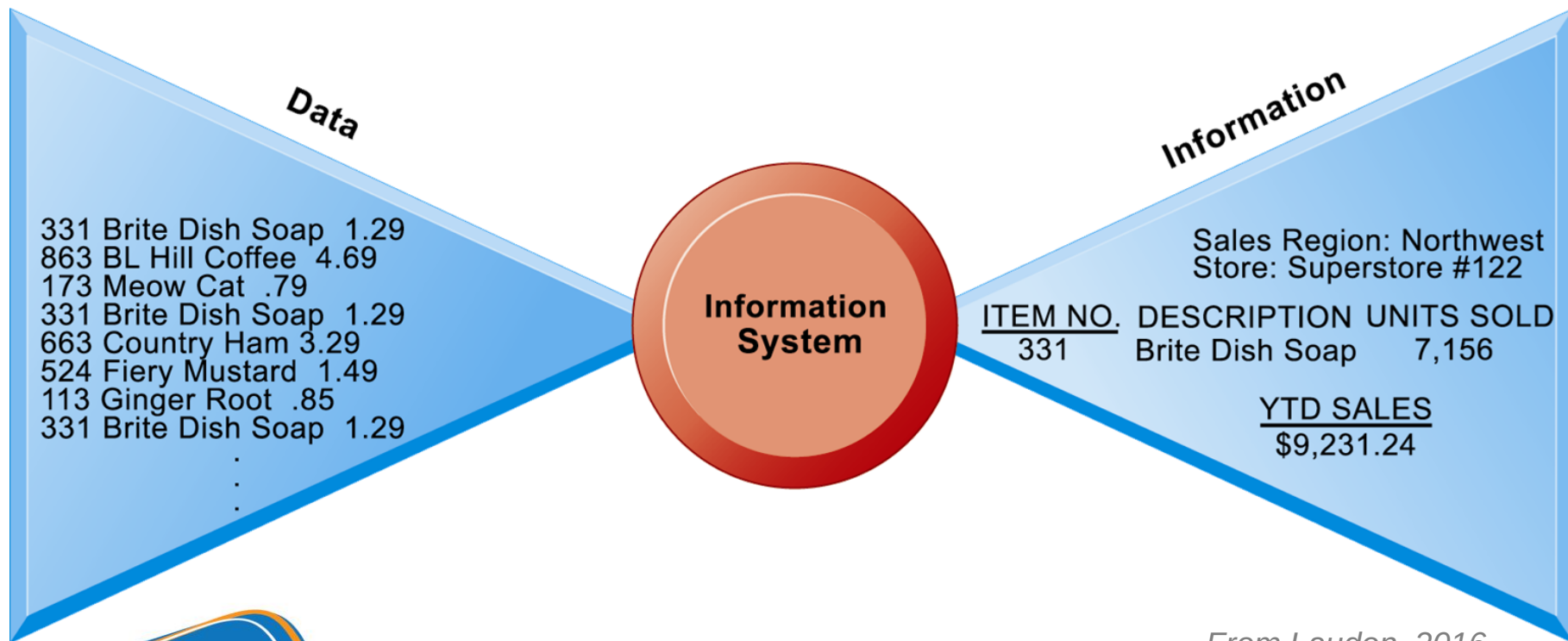
Information



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Knowledge

□ Example

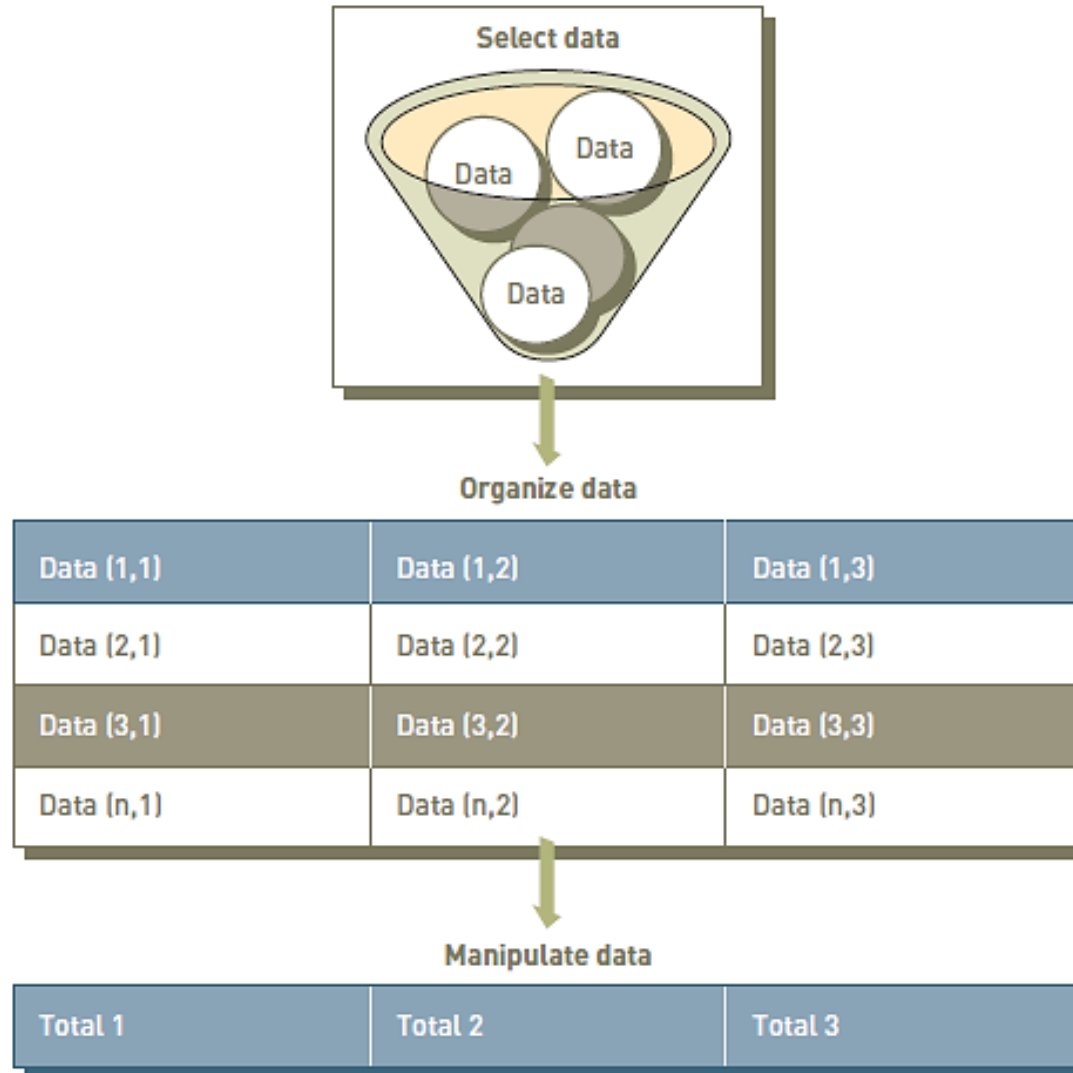


From Laudon, 2016

Data, Information, Knowledge

Data	Information	Knowledge
[No context]	[University context]	
3.16	3.16 + John Jones = GPA	* Job prospects
2.92	2.92 + Sue Smith = GPA	* Graduate school prospects
1.39	1.39 + Kyle Owens = GPA	* Scholarship prospects
3.95	3.95 + Tom Elias = GPA	
[No context]	[Professional baseball pitcher context]	
3.16	3.16 + Ken Rice = ERA	
2.92	2.92 + Ed Dyas = ERA	* Keep pitcher, trade pitcher, or send pitcher to minor leagues
1.39	1.39 + Hugh Carr = ERA	
3.95	3.95 + Nick Ford = ERA	* Salary/contract negotiations
GPA = grade point average (higher is better). ERA = earned run average (lower is better); ERA is the number of runs per nine innings that a pitcher surrenders.		

Process of Transforming Data into Information



- ☐ Which of the following is NOT an example of data?
- A. 231456
 - B. 32, CMT8, P6, Tan Binh
 - C. Blue
 - D. B, C

- ☐ Which of the following is NOT an example of information?
- A. 32, CMT8, P6, Tan Binh
 - B. Address
 - C. Address of my friend
 - D. B, C

- ☐ Which of the following is NOT an example of knowledge?
- A. 32, CMT8, P6, Tan Binh
 - B. Address
 - C. Address of my friend
 - D. Ability to find an address

- ☐ Which of the following is correct?
- A. Data leads to information, and information leads to knowledge.
 - B. Knowledge leads to information, and information leads to data
 - C. Data leads to knowledge and knowledge leads to information
 - D. A, B, C

- The value of information is directly linked to how it helps decision makers achieve their organization's goals
- Accuracy and completeness are critical for data used in accounting for the management of company assets, e.g., cash, inventory, and equipment

Characteristics of Valuable Information

Characteristics	Definitions
Accessible	Information should be easily accessible by authorized users so they can obtain it in the right format and at the right time to meet their needs.
Accurate	Accurate information is error free. In some cases, inaccurate information is generated because inaccurate data is fed into the transformation process. This is commonly called garbage in, garbage out (GIGO).
Complete	Complete information contains all the important facts. For example, an investment report that does not include all important costs is not complete.
Economical	Information should also be relatively economical to produce. Decision makers must always balance the value of information with the cost of producing it.
Flexible	Flexible information can be used for a variety of purposes. For example, information on how much inventory is on hand for a particular part can be used by a sales representative in closing a sale, by a production manager to determine whether more inventory is needed, and by a financial executive to determine the total value the company has invested in inventory.
Relevant	Relevant information is important to the decision maker. Information showing that lumber prices might drop might not be relevant to a computer chip manufacturer.

Characteristics of Valuable Information (cont'd.)

Characteristics	Definitions
Reliable	Reliable information can be trusted by users. In many cases, the reliability of the information depends on the reliability of the data-collection method. In other instances, reliability depends on the source of the information. A rumor from an unknown source that oil prices might go up might not be reliable.
Secure	Information should be secure from access by unauthorized users.
Simple	Information should be simple, not complex. Sophisticated and detailed information might not be needed. In fact, too much information can cause information overload, whereby a decision maker has too much information and is unable to determine what is really important.
Timely	Timely information is delivered when it is needed. Knowing last week's weather conditions will not help when trying to decide what coat to wear today.
Verifiable	Information should be verifiable. This means that you can check it to make sure it is correct, perhaps by checking many sources for the same information.

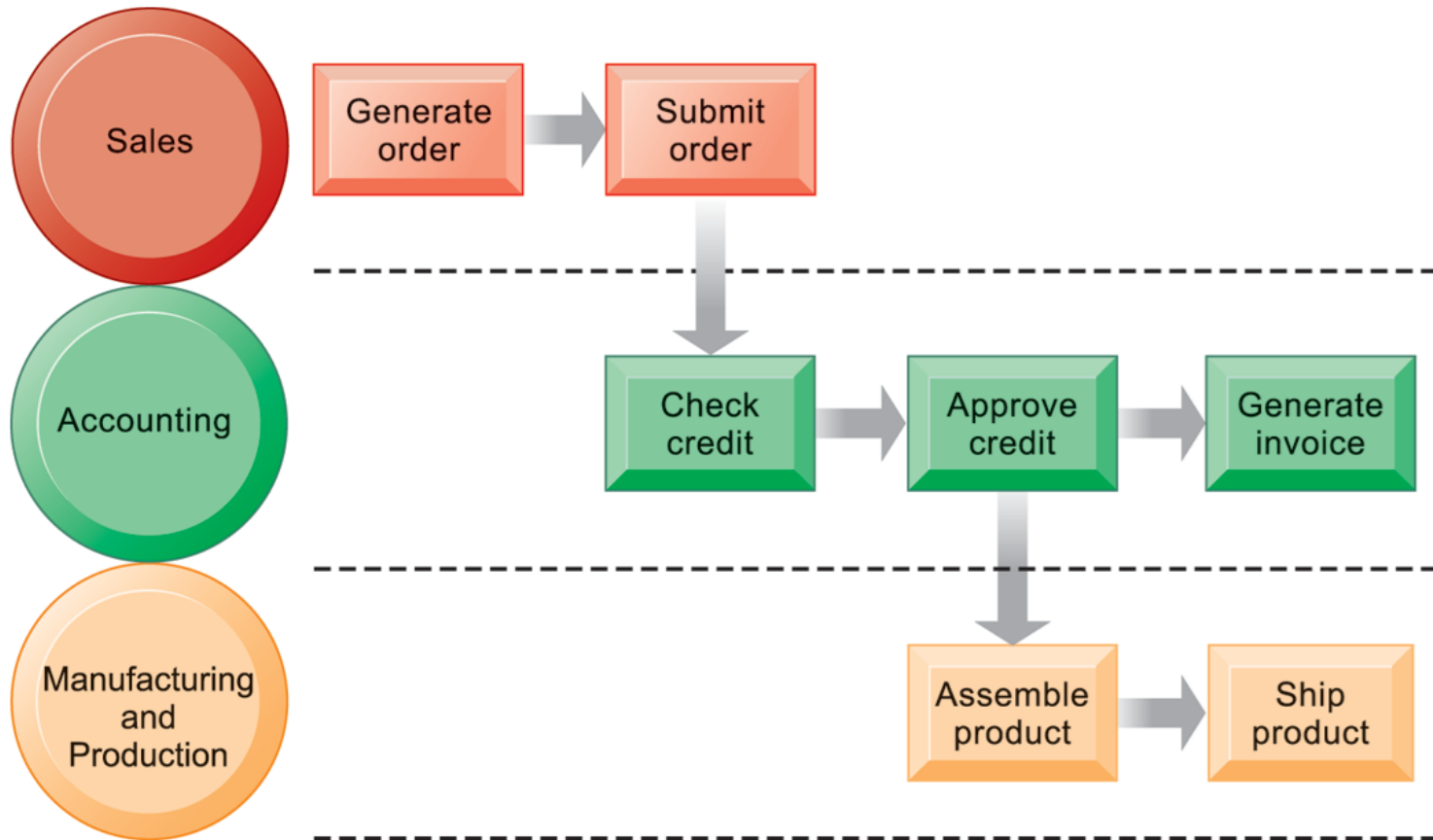
Business process

- The *manner* in which work is organized, coordinated, and focused to produce a valuable product/service
- The *collection of activities* required to produce a product/service
 - ▣ Supported by flows of material, information, and knowledge among the participants

Source: wikipedia

Example of a business process

□ The order fulfillment process



Information system development process

- Similar to software development process
 - ▣ Information systems are expressed through application software

- Software focuses on products, it specially focuses to the end-user factor and business
 - ▣ End-user: a person who performs tasks of a business process
 - ▣ Business: make value for business, improve performance

Information system development tools

- Tools support following stages
 - ▣ Requirement collection and analysis
 - ▣ Business process analysis
 - ▣ Data analysis
 - ▣ Information system analysis and design
 - ▣ Testing
 - ▣ Deployment
 - ▣ Operations/Maintenance

Components of an IS



Hardware



Software



Telecommunications

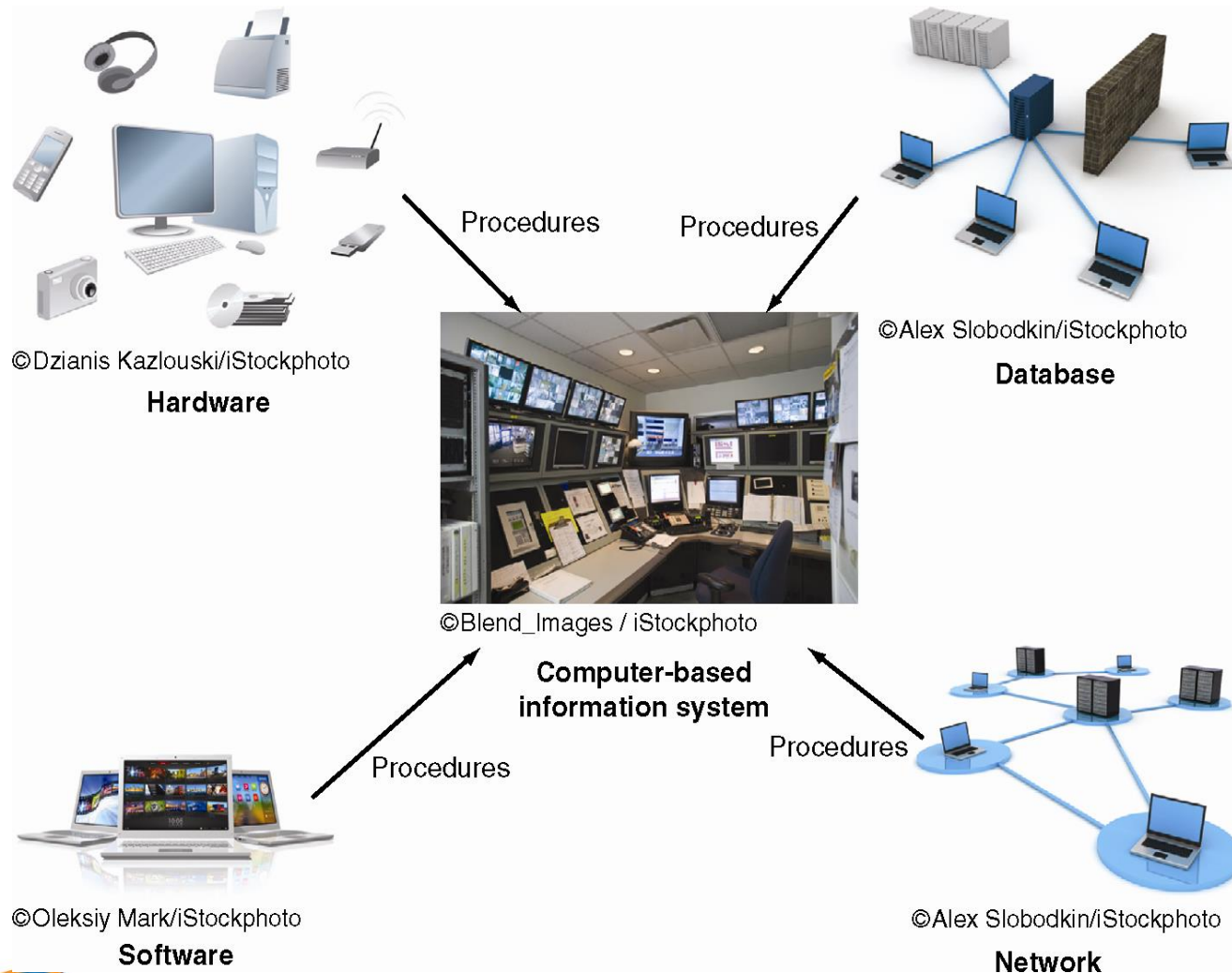


People



Procedures

Components of an IS



Hardware, Software, and Databases

- ☐ Hardware: computer equipment used to perform input, processing, storage, and output activities
- ☐ Software: computer programs that govern the operation of the computer
- ☐ Database: organized collection of facts and information
 - ☒ Typically consists of two or more related data files



Telecommunications, Networks, and the Internet

- Telecommunications: electronic transmission of signals for communications
- Networks connect computers and equipment enabling electronic communication
- Internet: world's largest computer network
 - ▣ Thousands of interconnected networks

Business Information System (BIS)

- A group of **interrelated components** that work collectively to carry out **input, processing, output, storage** and **control** actions in order to ***convert data into information products*** that can be used to support forecasting, control, coordination, decision making and operational activities in an organization.
- Most common types of information systems:
 - ▣ Systems for electronic and mobile commerce, transaction processing, management information, and decision support
- Special-purpose systems are utilized by organisations

Why BIS important?

☐ Operational Excellence

- ☐ Improve the efficiency of operations in order to achieve higher profitability. E.g. having the correct amount of stock in store so consumers can always get what they want

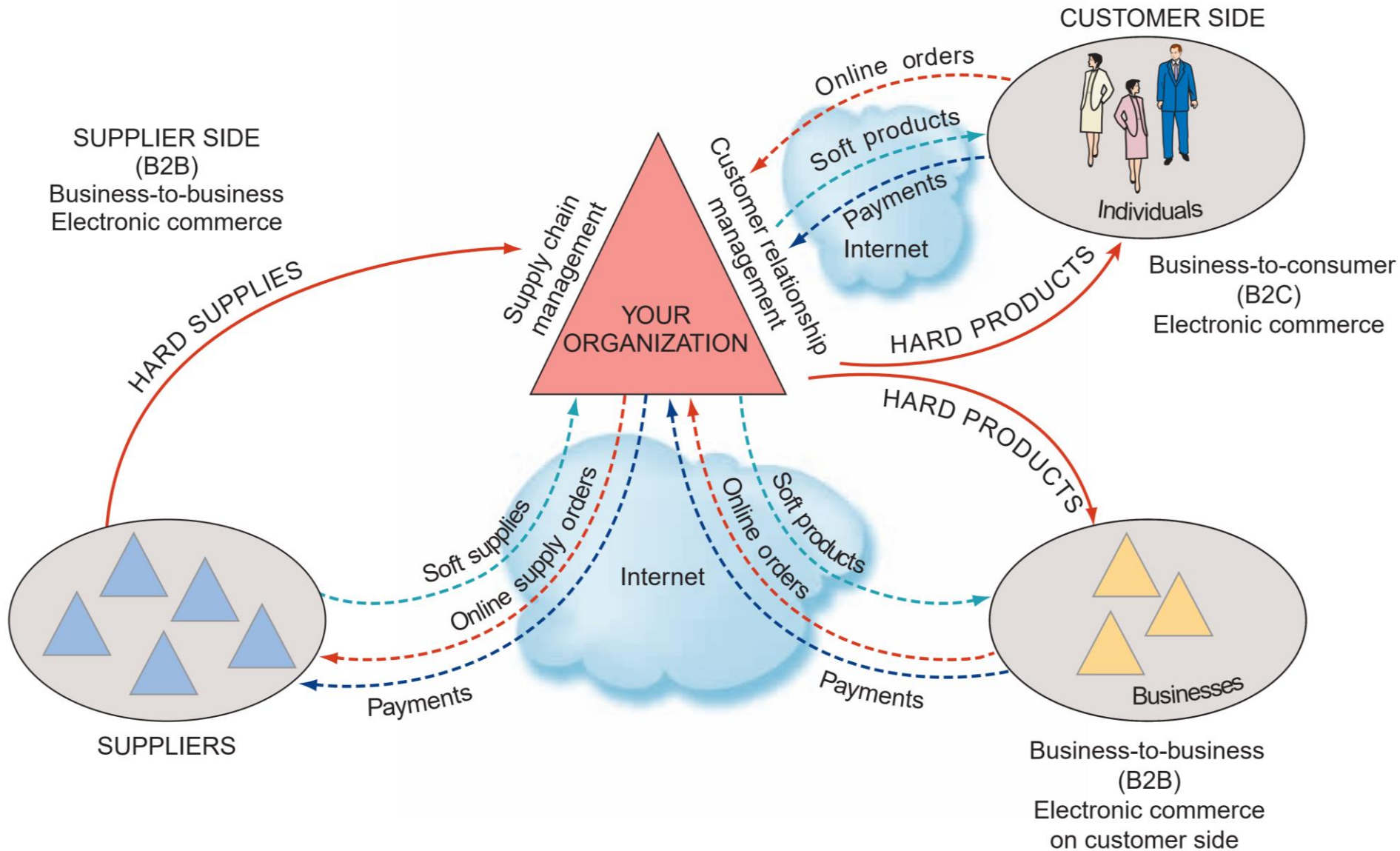
☐ New product services and business models

- ☐ IS systems play a major role for businesses in creating new products and services. New business models can be created and these can describe how a company produce, create and sell there products.

☐ Customer and Supplier intimacy

- ☐ The better services a company provides its consumers with more likely they are too come back to them and as result the more they will buy off the supplier therefore creating a good relationship with both parties.

BIS - Example



Why BIS important?

- Improved decision making
 - ▣ Managers can use real time data to make better decisions and not have to waste time looking for information.
- Competitive advantage
 - ▣ IS systems will generally create a competitive advantage over their rivals
- Day to Day survival
 - ▣ Business invest in these systems to make their jobs as easy as possibly.
 - ▣ Example: is Citibank introduced the first ATM machine to make it easier for customers to access their money and to cut down queues in their banks.

Major branches of BIS

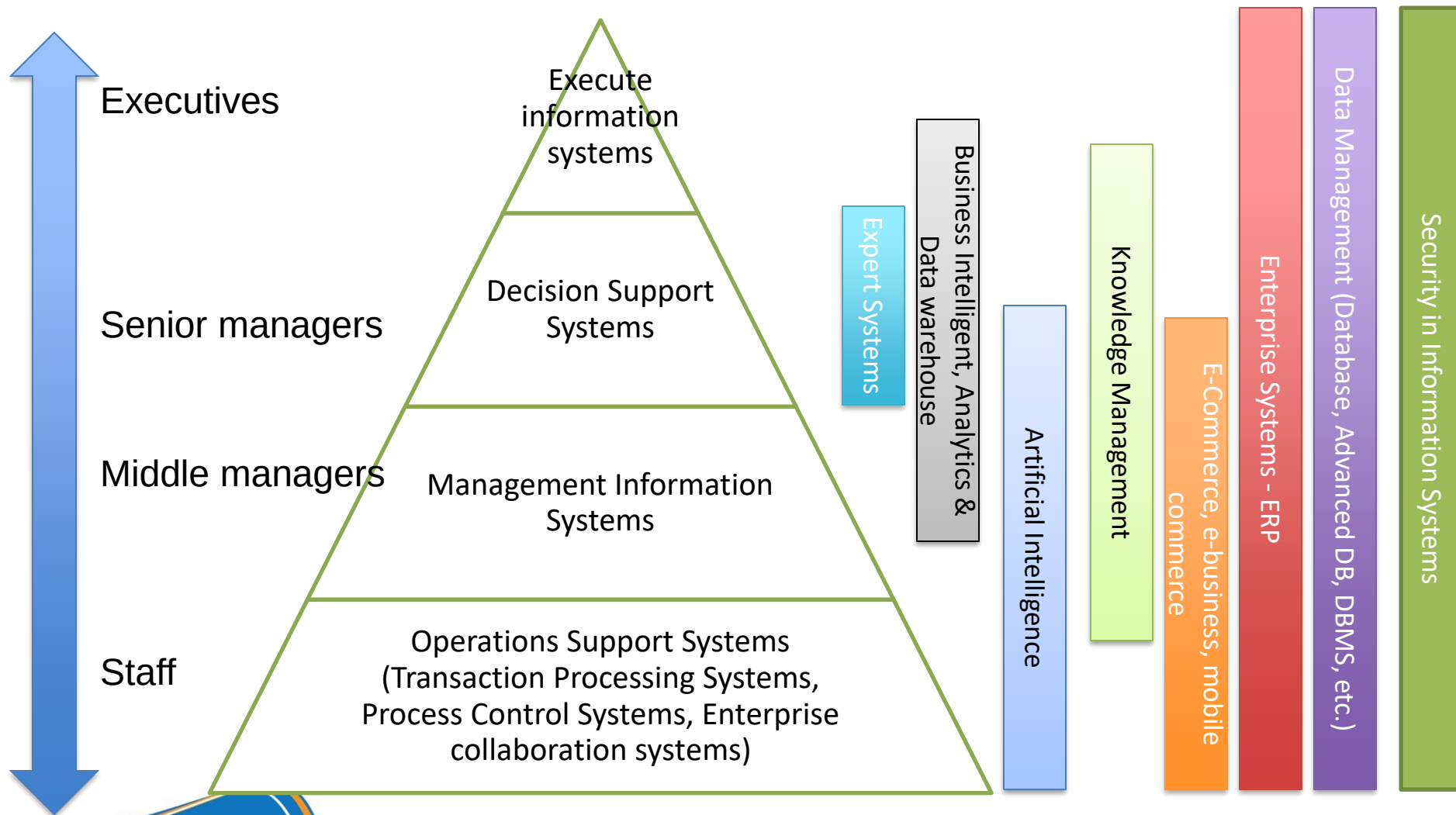




Type of BIS

Type of System	Function	Example
Functional area IS	Supports the activities within specific functional area.	System for processing payroll
Transaction processing system	Processes transaction data from business events.	Walmart checkout point-of-sale terminal
Enterprise resource planning	Integrates all functional areas of the organization.	Oracle, SAP system
Office automation system	Supports daily work activities of individuals and groups.	Microsoft® Office
Management information system	Produces reports summarized from transaction data, usually in one functional area.	Report on total sales for each customer
Decision support system	Provides access to data and analysis tools.	“What-if” analysis of changes in budget
Expert system	Mimics human expert in a particular area and makes decisions.	Credit card approval analysis
Executive dashboard	Presents structured, summarized information about aspects of business important to executives.	Status of sales by product
Supply chain management system	Manages flows of products, services, and information among organizations.	Walmart Retail Link system connecting suppliers to Walmart
Electronic commerce system	Enables transactions among organizations and between organizations and customers.	www.dell.com

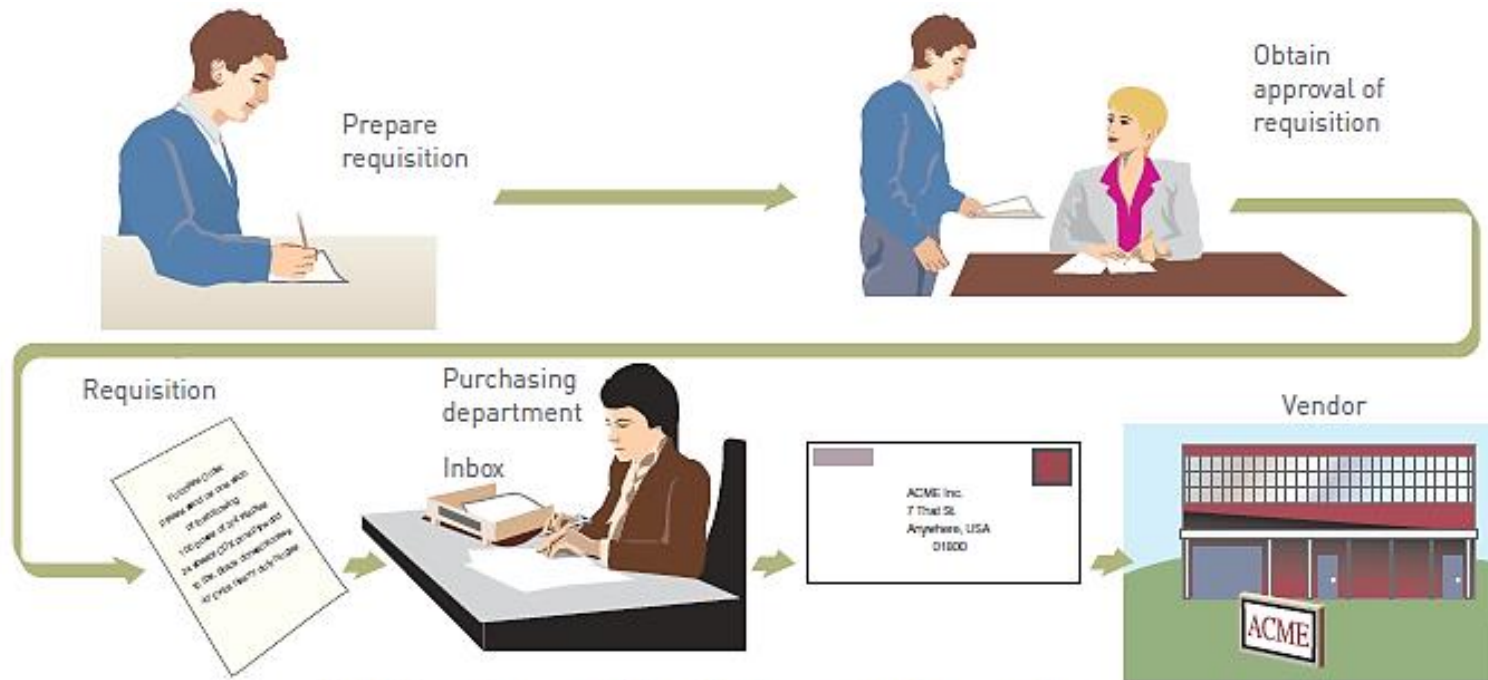
Major branches of BIS



- E-commerce: business transactions executed electronically
 - ▣ Business-to-business (B2B)
 - ▣ Business-to-consumer (B2C)
 - ▣ Consumer-to-consumer (C2C)
 - ▣ Between business and the public sector
 - ▣ Between consumers and the public sector

- ☐ Mobile commerce (m-commerce): the use of mobile, wireless devices to place orders and conduct business
- ☐ E-commerce offers many advantages for streamlining work activities
- ☐ Electronic business (e-business) uses information systems and the Internet to perform all business-related tasks and functions

Electronic Commerce



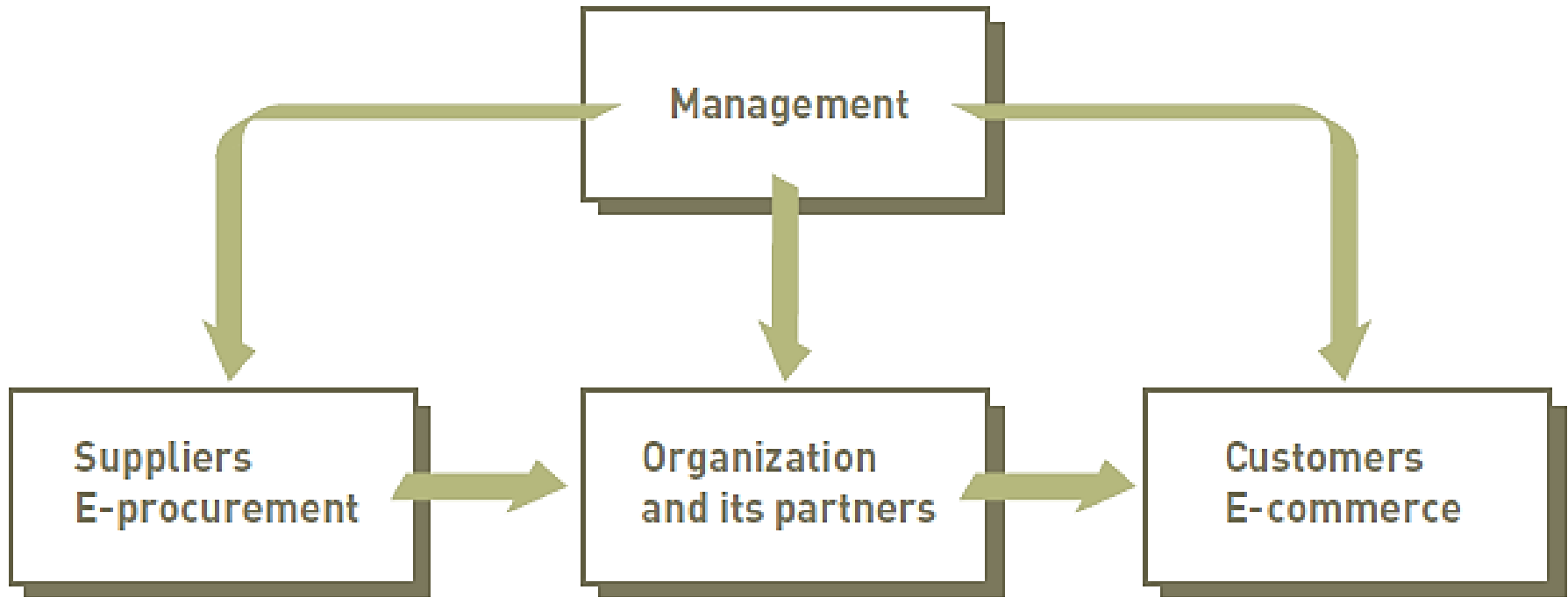
Traditional process for placing a purchase order



E-commerce process for placing a purchase order

Electronic Business (E-business)

E-BUSINESS



Enterprise Systems- Transaction Processing Systems

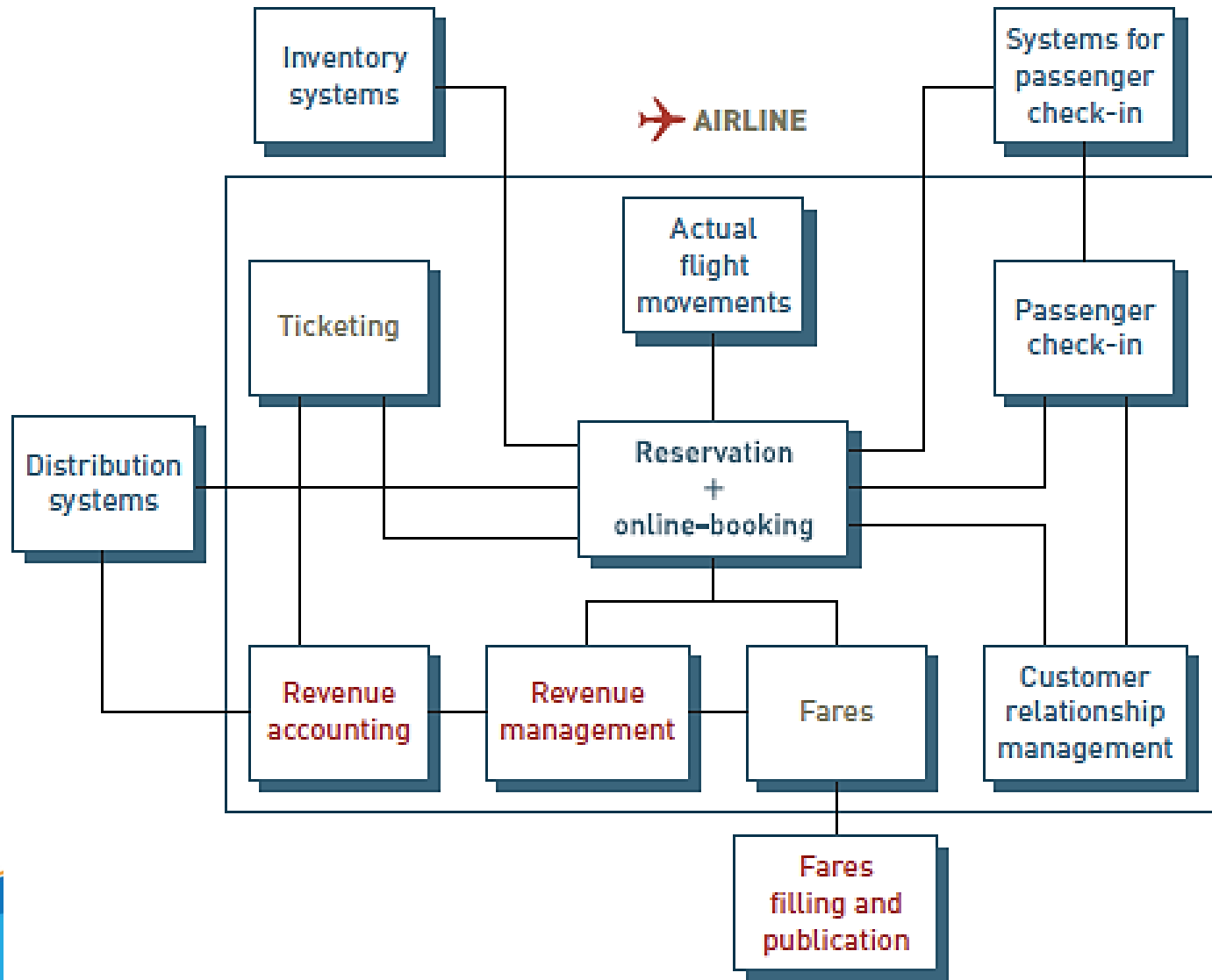
□ Transaction

- Any business-related exchange, such as payments to employees and sales to customers

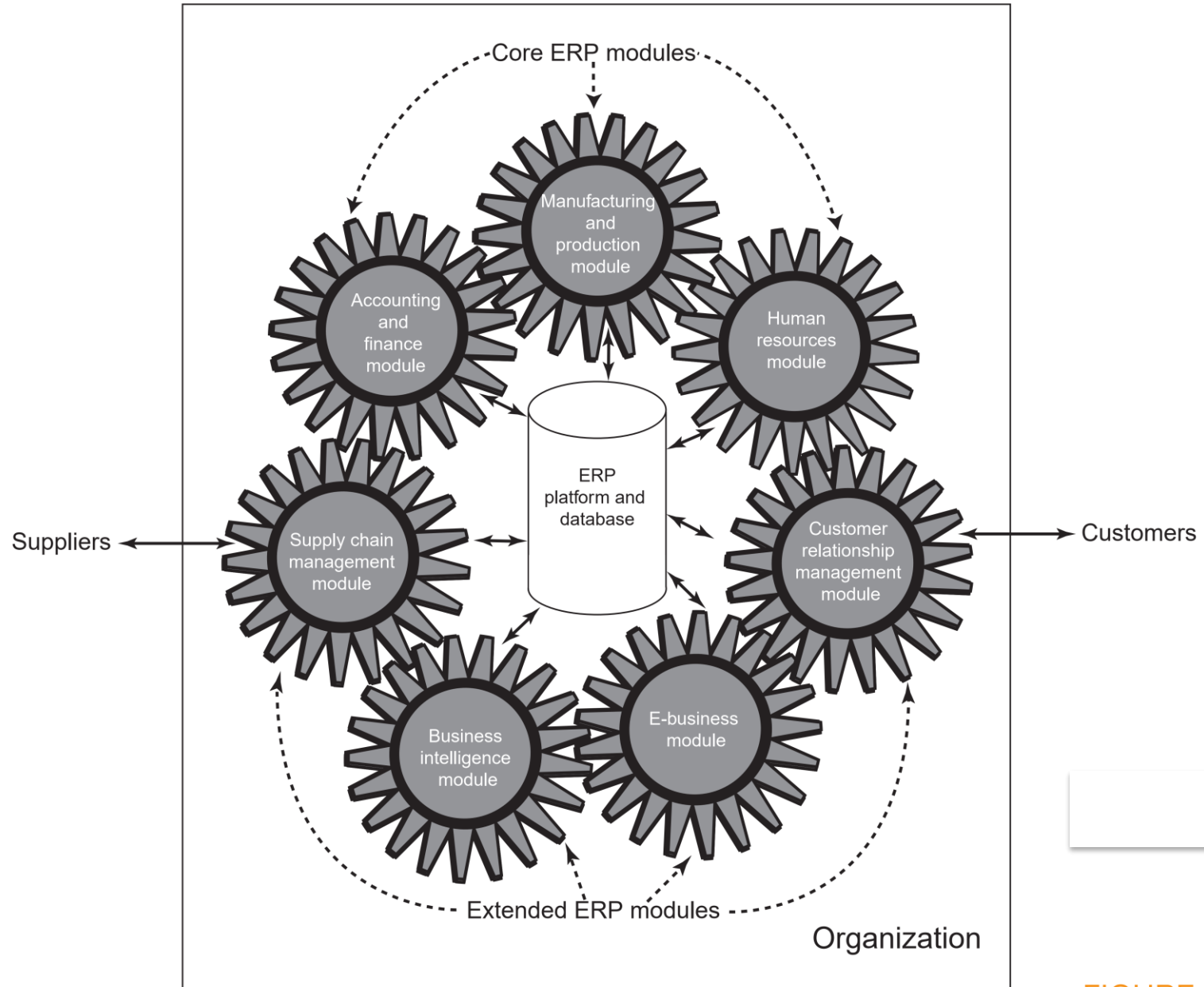
□ Transaction processing system (TPS)

- An organized collection of people, procedures, software, databases, and devices
- Used to perform and record completed business transactions

Integrated Transaction Processing System

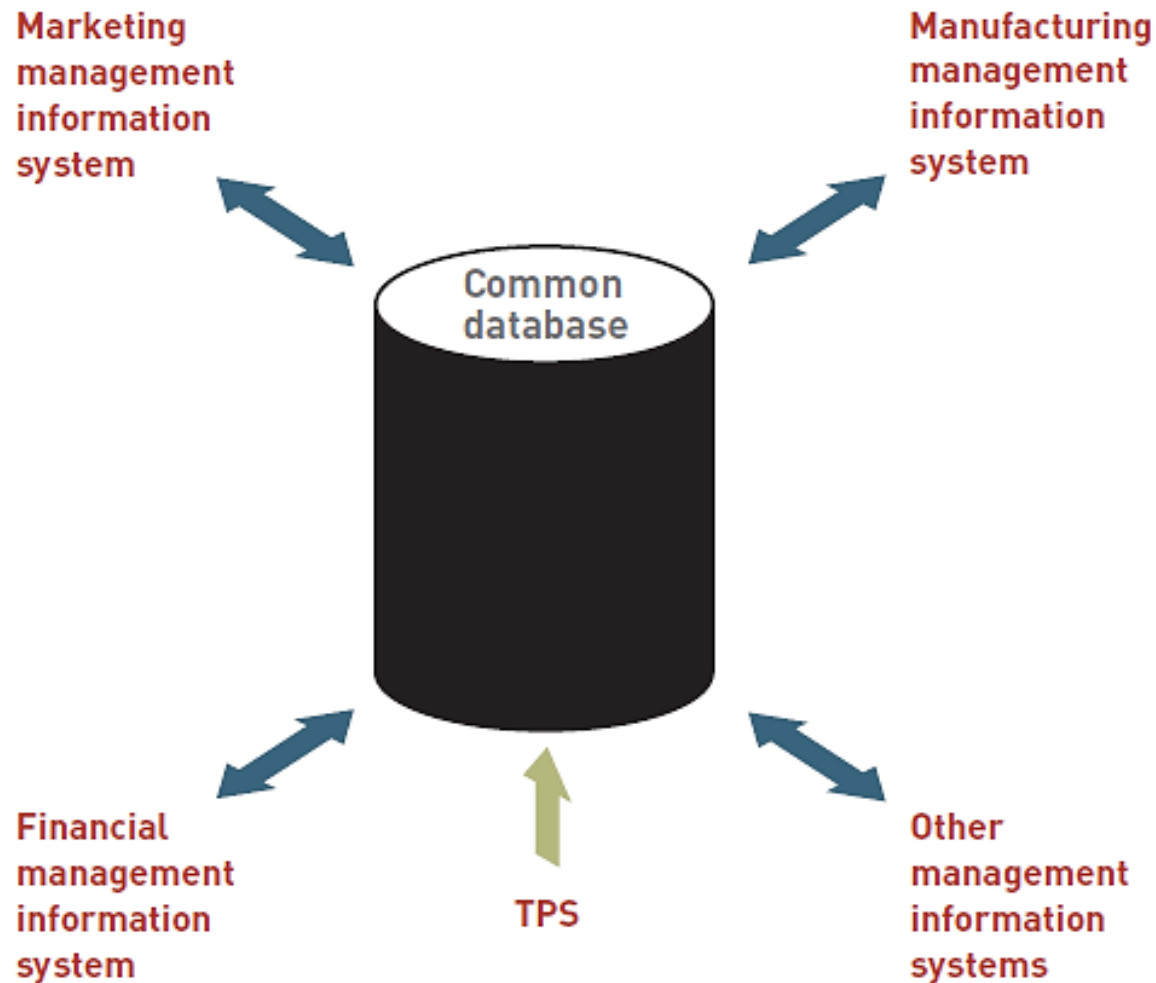


- Enterprise resource planning (ERP) system
 - ▣ A set of integrated programs
 - ▣ Manages the vital business operations for an entire multisite, global organization
- Most ERP systems provide integrated software to support manufacturing and finance



- Management information system (MIS)
 - Organized collection of people, procedures, software, databases, and devices
 - Provides routine information to managers and decision makers
 - Focuses on operational efficiency
 - Provides standard reports generated with data and information from the TPS or ERP

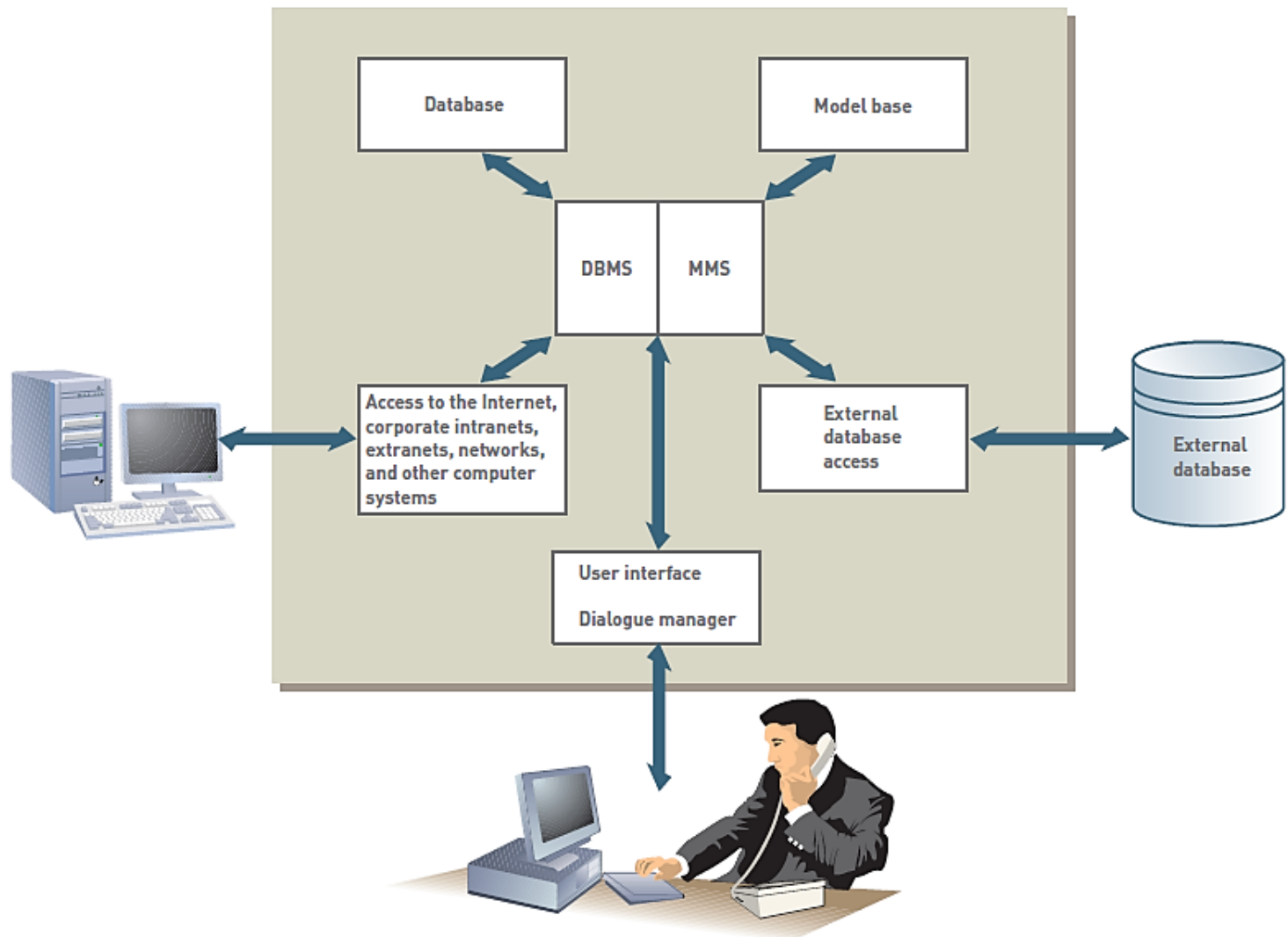
Management Information System



Decision Support System (DSS)

- An organized collection of people, procedures, software, databases, and devices that support problem-specific decision making
 - ▣ Focus is on making effective decisions
- A DSS can include the following:
 - ▣ Model base
 - ▣ Database
 - ▣ User interface or dialogue manager

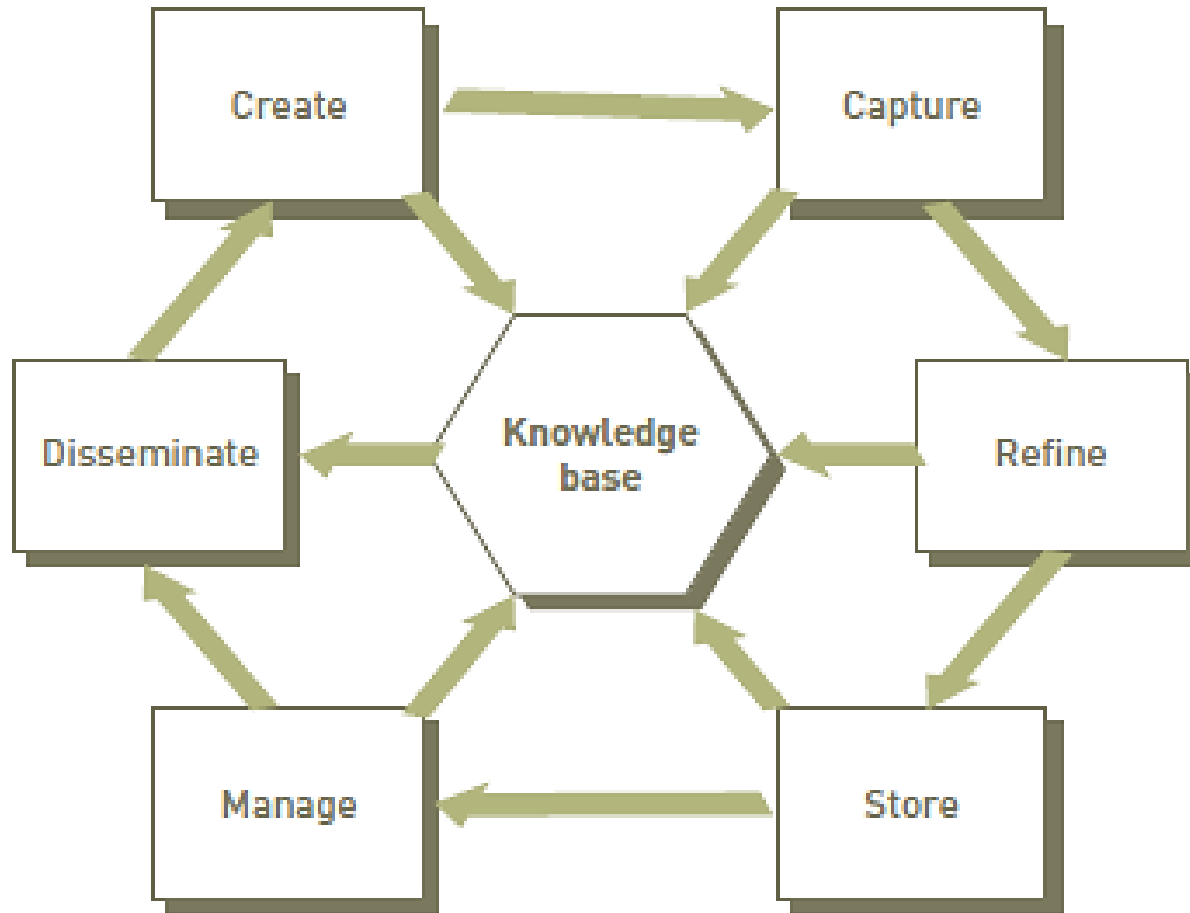
Essential DSS Elements



Knowledge Management

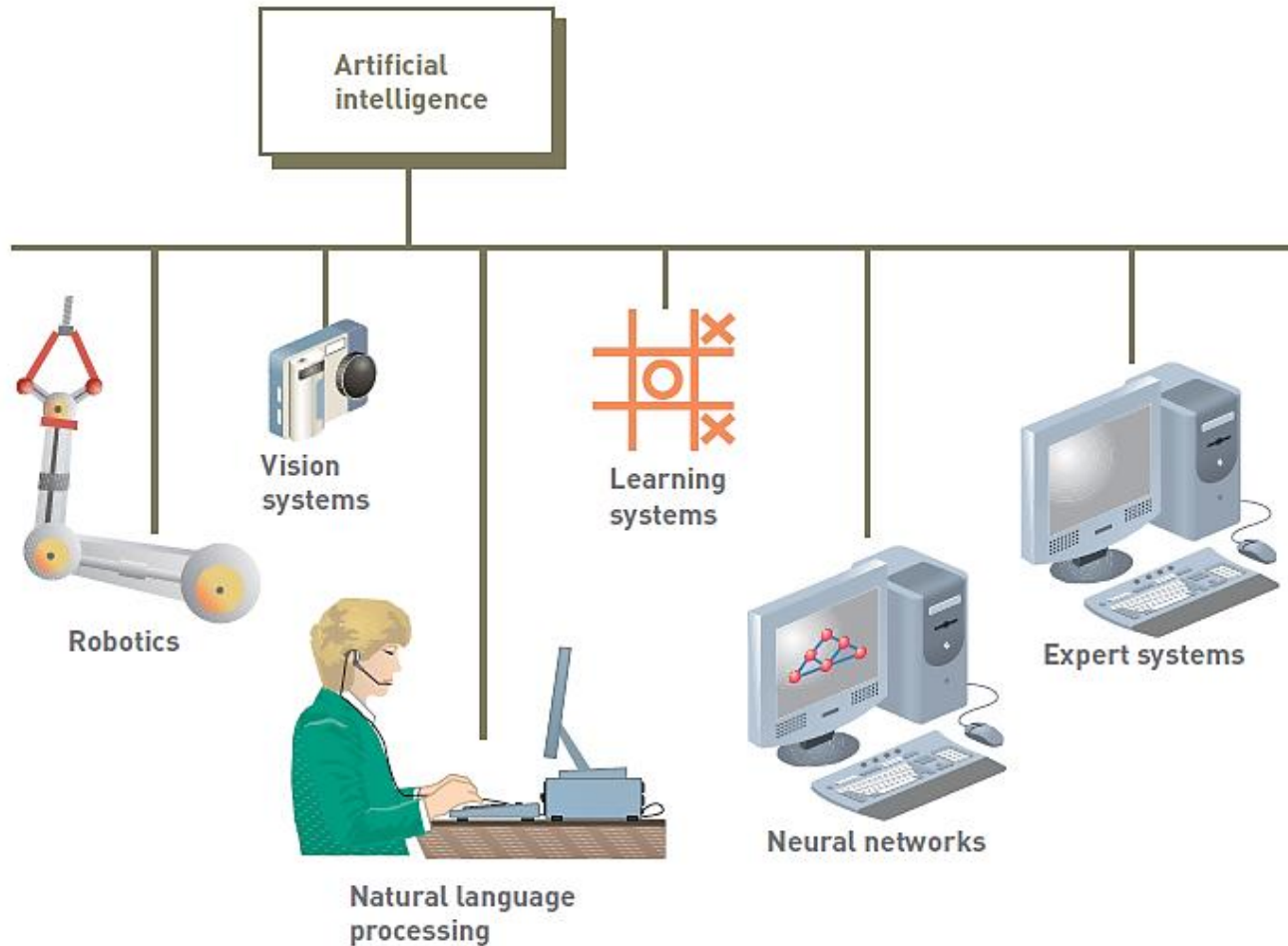
- Knowledge management systems (KMSs)
 - An organized collection of people, procedures, software, databases, and devices
 - Create, store, share, and use the organization's knowledge and experience

Knowledge Management Process



- A computer system takes on characteristics of human intelligence
- Examples of AI applications: games, medical diagnoses, and automobile operation

Major Branches of Artificial Intelligence



Artificial Intelligence Subfields

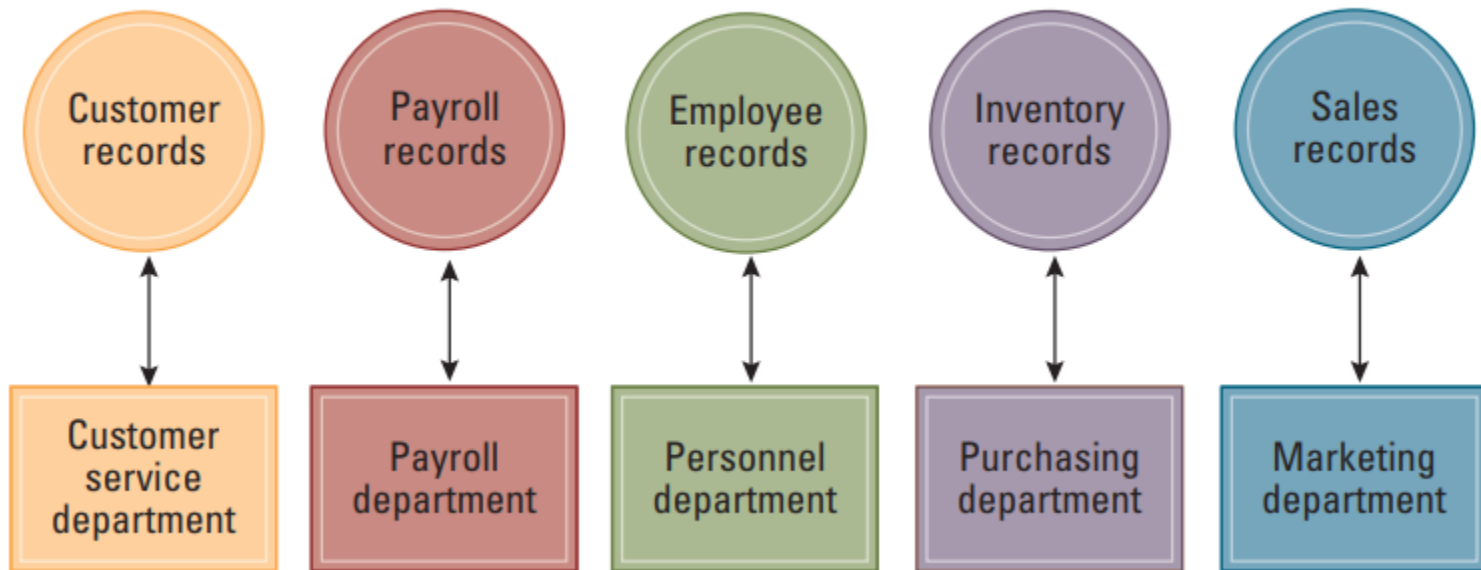
- ☐ Robotics: machines take over complex, dangerous, routine or boring tasks
- ☐ Vision systems allow devices to see, store and process images
- ☐ Natural language processing involves computers understanding and acting on verbal or written commands
- ☐ Learning systems allow computers to learn from past mistakes or experiences
- ☐ Neural networks allow computers to recognize and act on patterns or trends

- Expert systems give computers the ability to make suggestions and function like an expert in a particular field
 - ▣ Knowledge base contains data, rules, procedures, and relationships used by expert system
- Virtual reality: simulation of a real or imagined environment that can be experienced visually in three dimensions

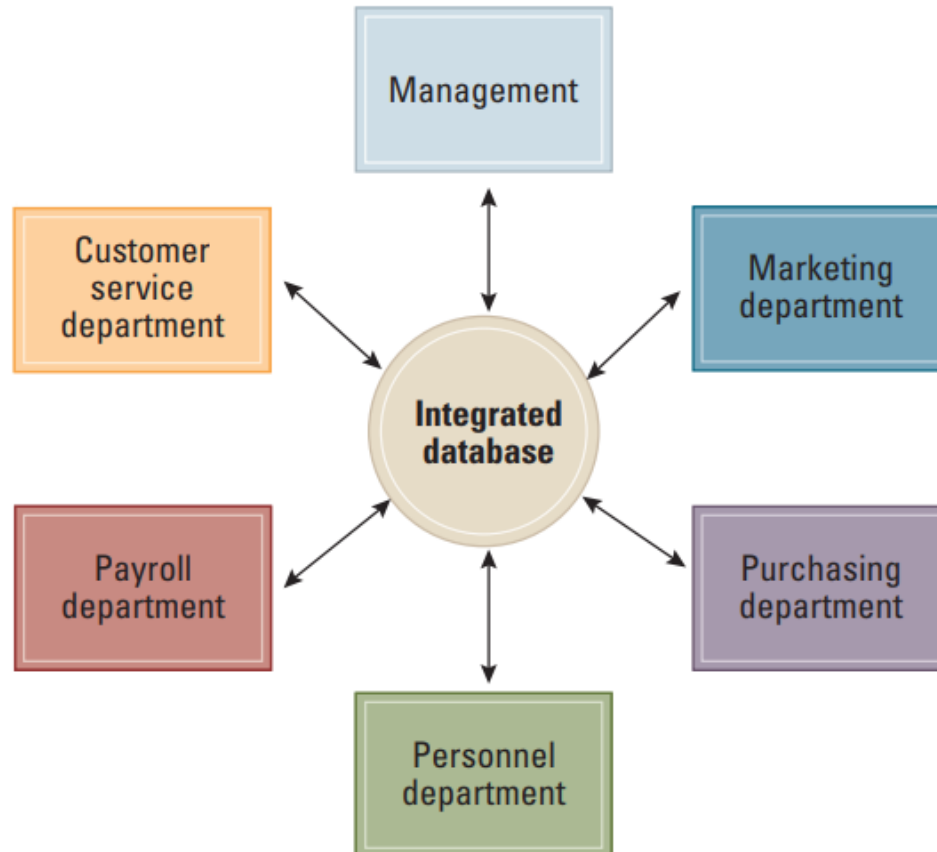
Database Fundamentals

- Flat file: one-dimensional storage system, meaning that it presents its information from a single point of view

a. File-oriented information system



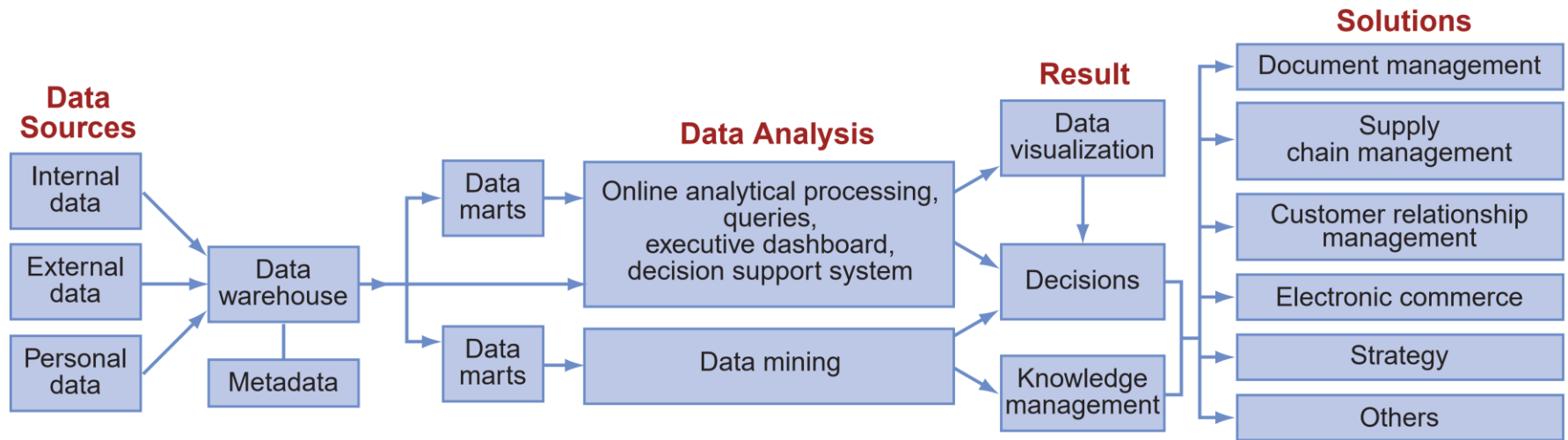
- Database refers to a **collection of data** that is multidimensional in the sense that internal links between its entries make the information accessible from a variety of perspectives



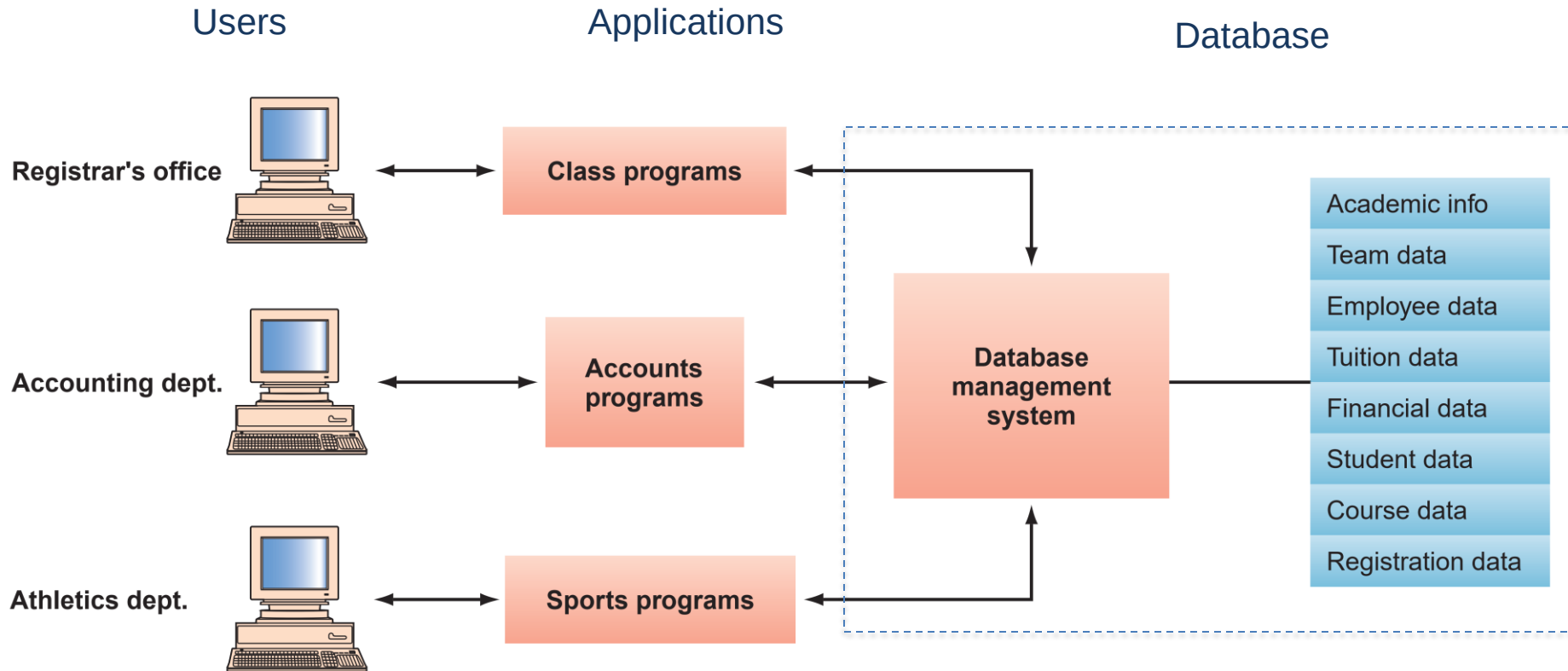
- A **schema** is a description of the entire **database structure** that is used by the database software to maintain the database.
- A **subschemas** is a description of only that **portion** of the database pertinent to a particular user's needs

- A schema for a university database would indicate that:
 - ▣ Each student record contains such items as the current address and phone number of that student in addition to the student's academic record.
 - ▣ Each student record is linked to the record of the student's faculty adviser.
 - ▣ In turn, the record for each faculty member would contain the person's address, employment history, and so on.
 - ▣ Based on this schema, a linkage system would be maintained

Data Life Cycle



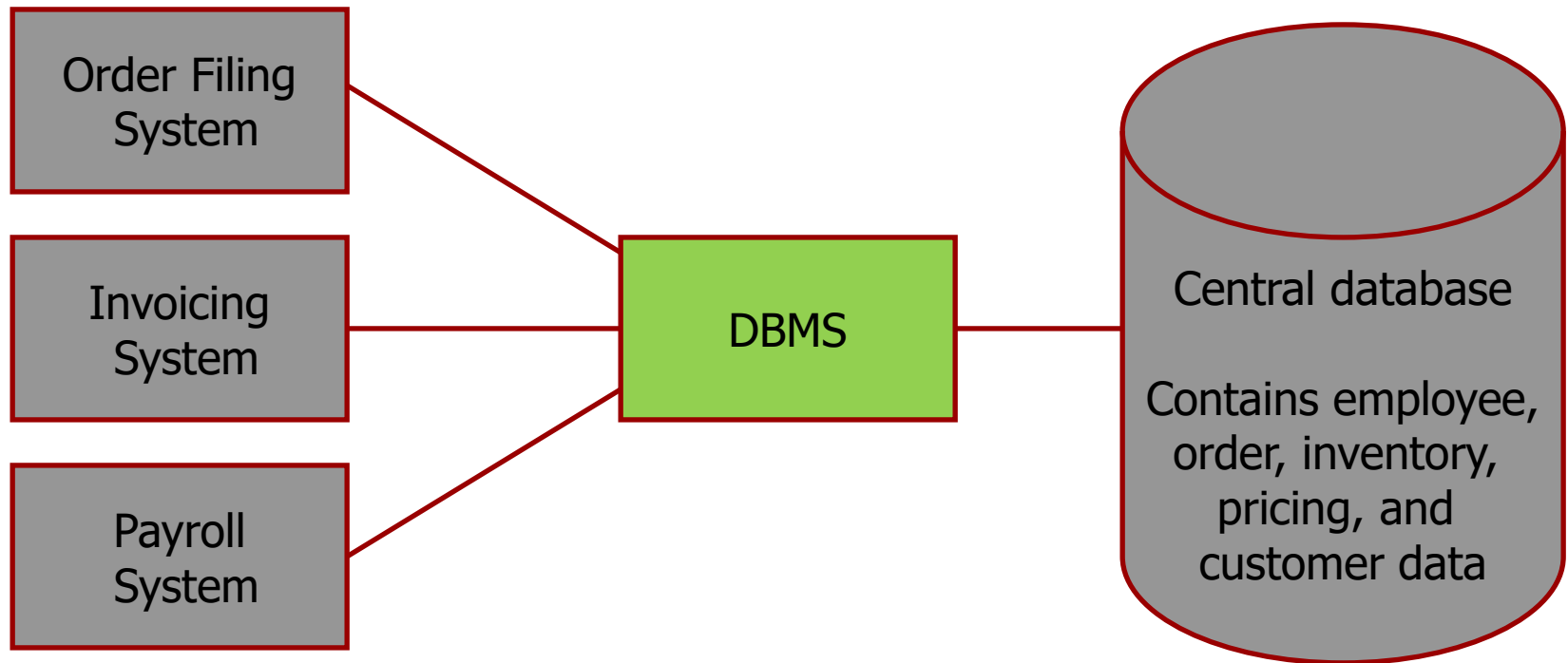
Database Approach



A database management system (DBMS) provides access to all data in the database.

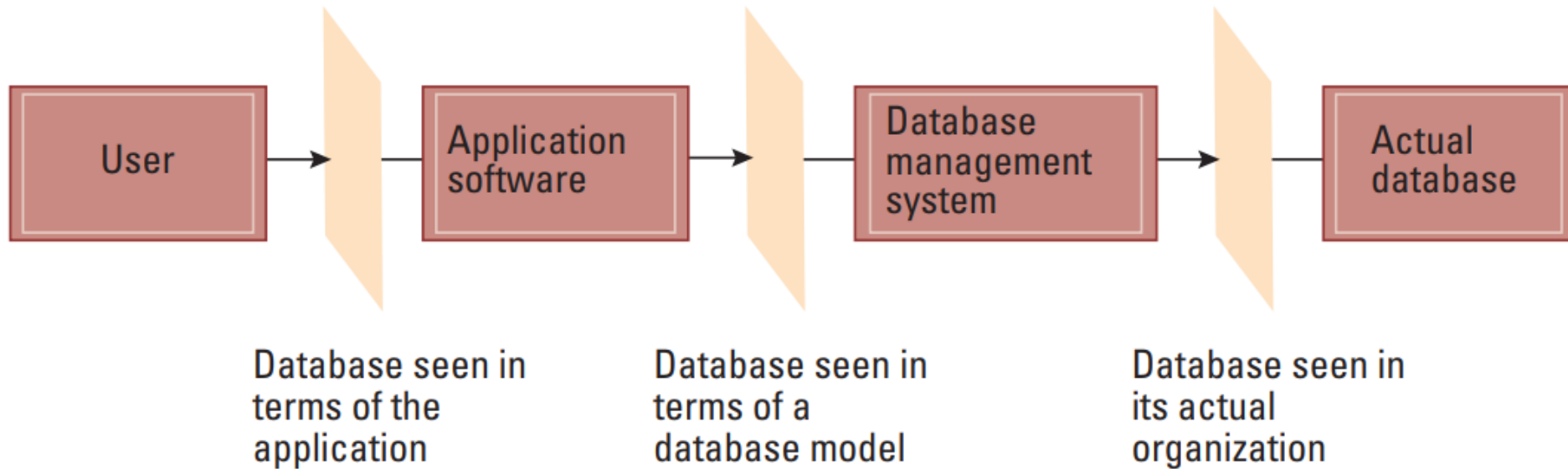
Database Management System

A software system that is used to create, maintain, and provide controlled access to user databases



DBMS manages data resources like an operating system manages hardware resources

Conceptual layers of a database implementation



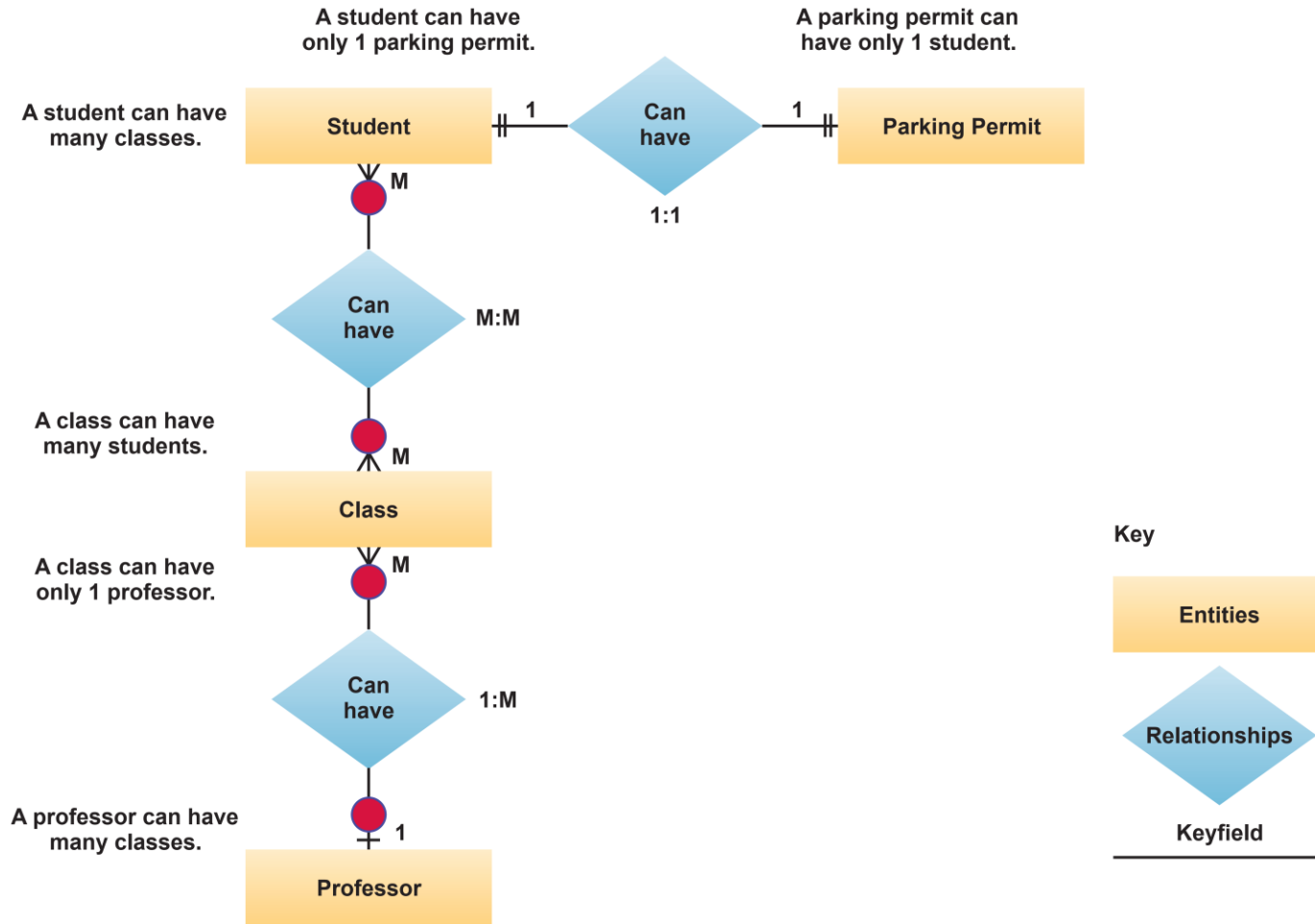
From Brookshear, 2015

Database models

- A conceptual view of a database
 - Entity Relationship Model (ERM)
 - Relational Model
 - Object-oriented Model
 - Hierarchical Model
 - NoSQL
 - ...

- ERM, Relational model is the most popular model

Example: Entity Relationship Model



(a) ER diagram

Example: Relational database model

CUSTOMER

<u>CustomerID</u>	CustomerName	CustomerAddress	CustomerCity*	CustomerState*	CustomerPostalCode
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Primary Key

ORDER

<u>OrderID</u>	OrderDate	<u>CustomerID</u>
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Foreign Key

(implements 1:N relationship between customer and order)

ORDER LINE

<u>OrderID</u>	<u>ProductID</u>	OrderedQuantity
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Combined, these are a *composite primary key* (uniquely identifies the order line)...individually they are *foreign keys* (implement M:N relationship between order and product)

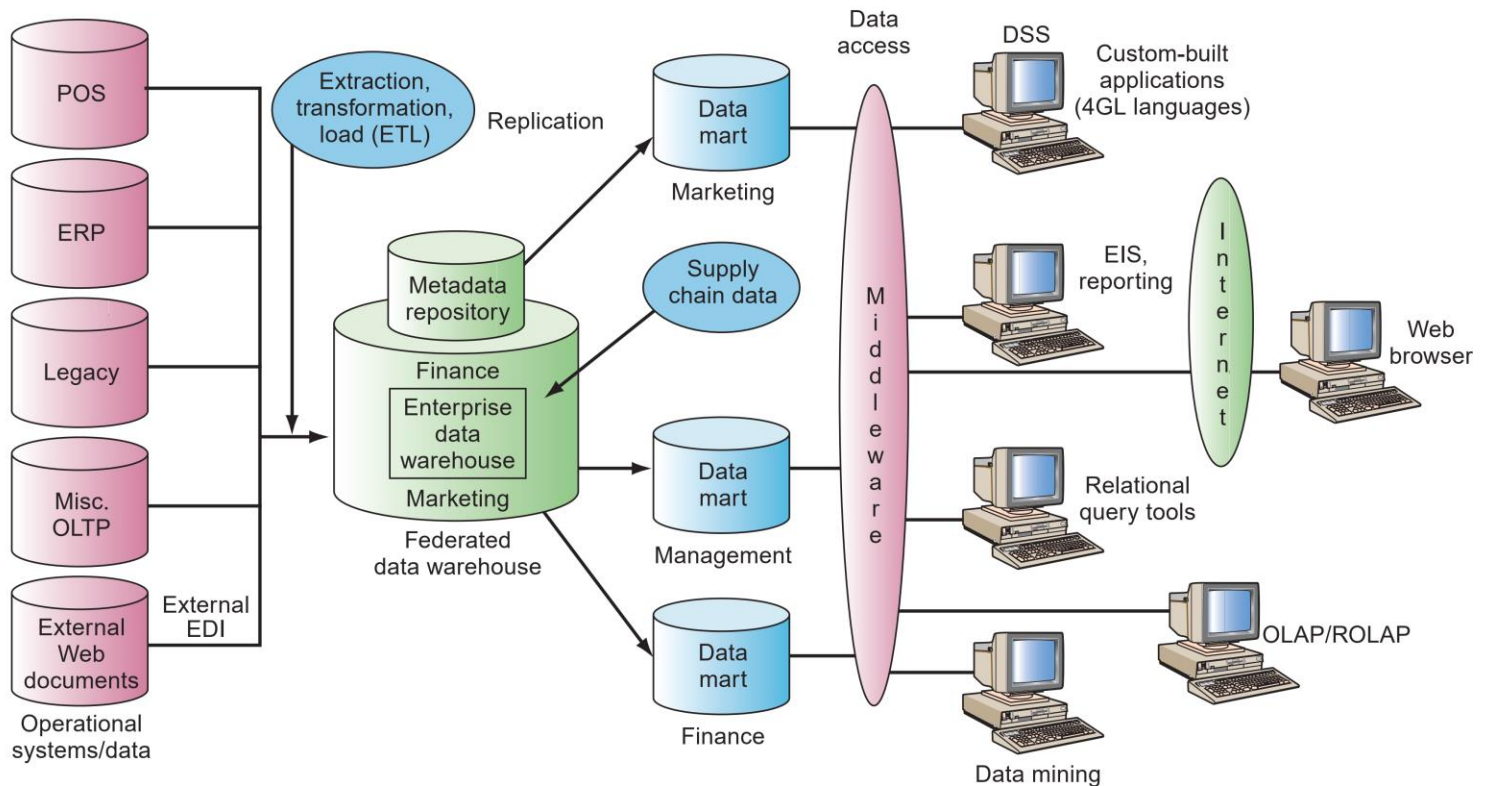
PRODUCT

<u>ProductID</u>	ProductDescription	ProductFinish	ProductStandardPrice	ProductLineID
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Datawarehouse

Data warehouse (DW or DWH), also known as an enterprise data warehouse (EDW), is a system used for reporting and data analysis, and is considered a core component of business intelligence. DWs are central repositories of integrated data from one or more disparate sources. (Wikipedia)

Business intelligence



Data warehouse framework and views

Careers in Information Systems

□ Successful IS workers must:

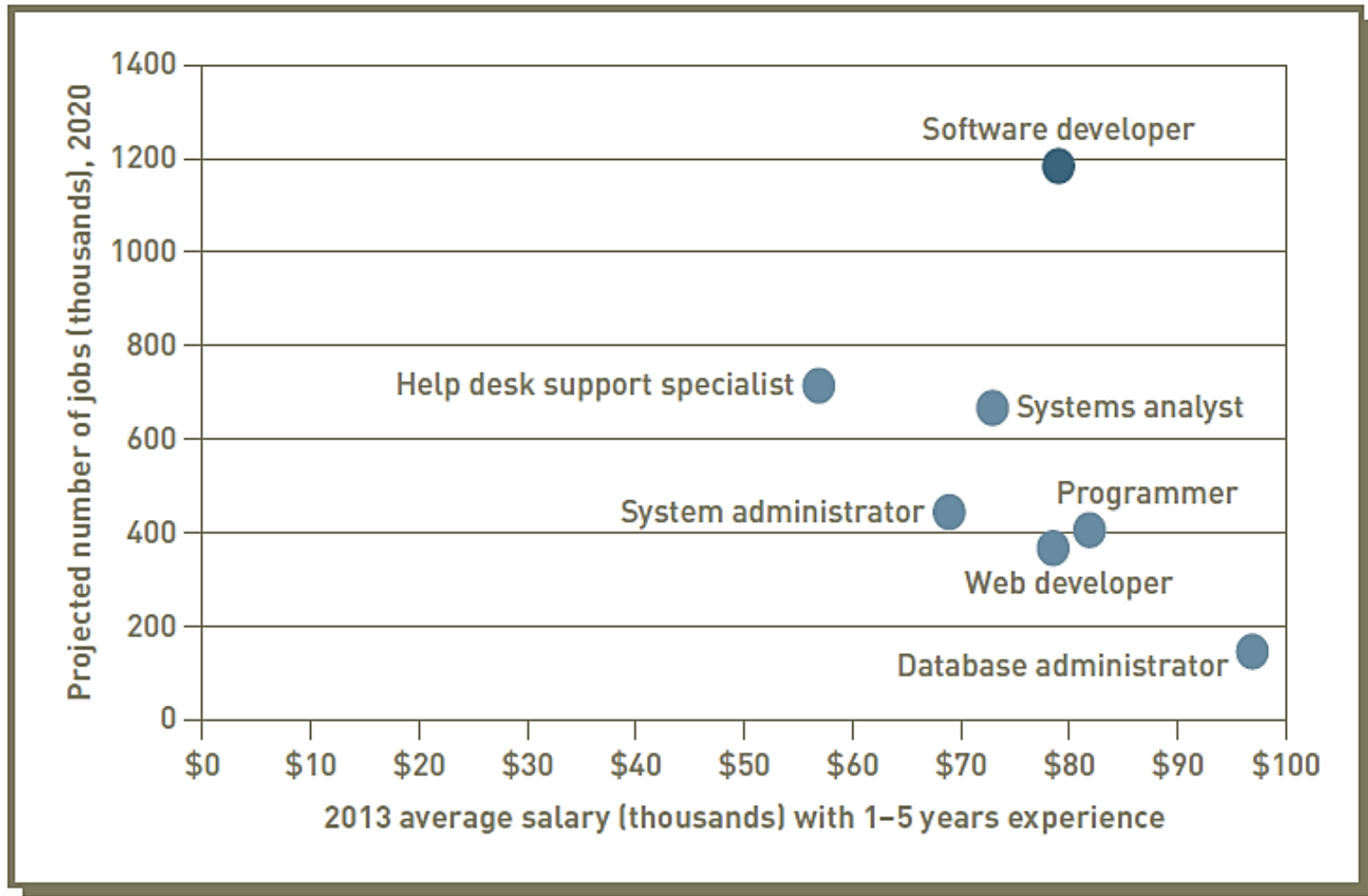
- Enjoy working in a fast-paced, dynamic environment
- Meet deadlines and solving unexpected challenges
- Possess good communication skills
- Have solid analytical and decision-making skills
- Develop effective team and leadership skills
- Be adept at implementing organization change
- Be prepared to engage in life-long learning



Important Skills for IS Workers

- ☐ Mobile applications for smartphones, tablet computers, and other mobile devices
- ☐ Programing and application development
- ☐ Help desk and technical support
- ☐ Project management
- ☐ Networking
- ☐ Business intelligence
- ☐ Security
- ☐ Web 2.0, 3.0
- ☐ Data center

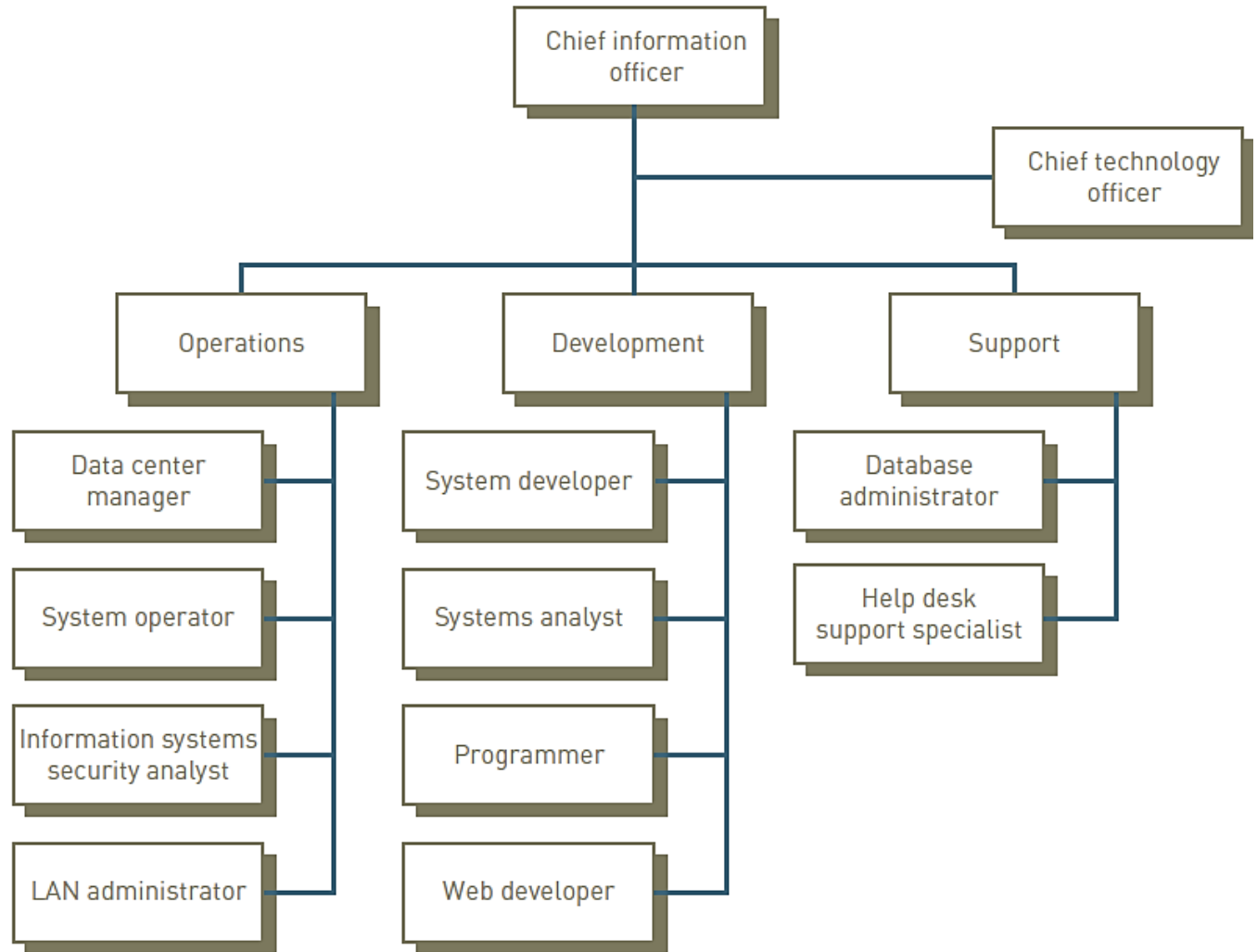
Occupational Outlook for Selected Information Systems Positions



Roles, Functions, and Careers in IS

- In addition to technical skills, IS professionals need:
 - Communication: skills in written and verbal communication
 - Problem solving
 - An understanding of organizations and business and the way they operate
 - Team work: ability to work with people and in groups
 - Etc.

Three Primary Functions of the Information Systems Organization



Typical IS Titles and Functions

- Chief Information Officer (CIO) employs the IS department's equipment and personnel to help the organization attain its goals
- Senior IS Managers
 - IS Director
 - Vice President of Information Systems
 - Manager of Information Systems
 - Chief Technology Officer (CTO)
 - Information Management Manager

Typical IS Titles and Functions: Operations Roles

- Data center managers:
 - ▣ Responsible for the maintenance and operation of the organization's computing facilities
- Operations group includes:
 - ▣ Operations Manager
 - ▣ System Operators
 - ▣ Information System Security Analysts



Typical IS Titles and Functions: Development Roles

- ☐ The development group implements the new information systems required to support the organization's existing and future business
- ☐ The development group includes:
 - ☐ IS Project Manager
 - ☐ Systems Analyst (SA), Business Analyst (BA)
 - ☐ Database designer, Web designer
 - ☐ Programmers
 - ☐ Web developers



Typical IS Titles and Functions:

Support

- ☐ Provides customer service for the employees, customers, and business partners

- ☐ The support group includes:
 - ☐ Database Administrators (DBAs)
 - ☐ Help Desk Support Specialists



IS Careers in Technology Organizations

- ☐ Consulting opportunities
- ☐ Computer training
- ☐ Computer and computer-equipment sales
- ☐ Computer repair and maintenance
- ☐ Many other related careers

Information Systems Majors – IS Department



- Room 184 – Phone 38354266 (804)
- Head: Dr. Pham Nguyen Cuong
- Vice head: Dr. Nguyen Tran Minh Thu

Educational objectives

- ☐ Be able to ***analyze, evaluate, design, implement and operate*** projects of information system
- ☐ Be able to ***utilize and self-study*** Information System development tools (*analysis and design, software development via network / distributed / mobile environment*)
- ☐ Have ***self-development skills, do research*** of technologies, methodologies, new processes in Information System area

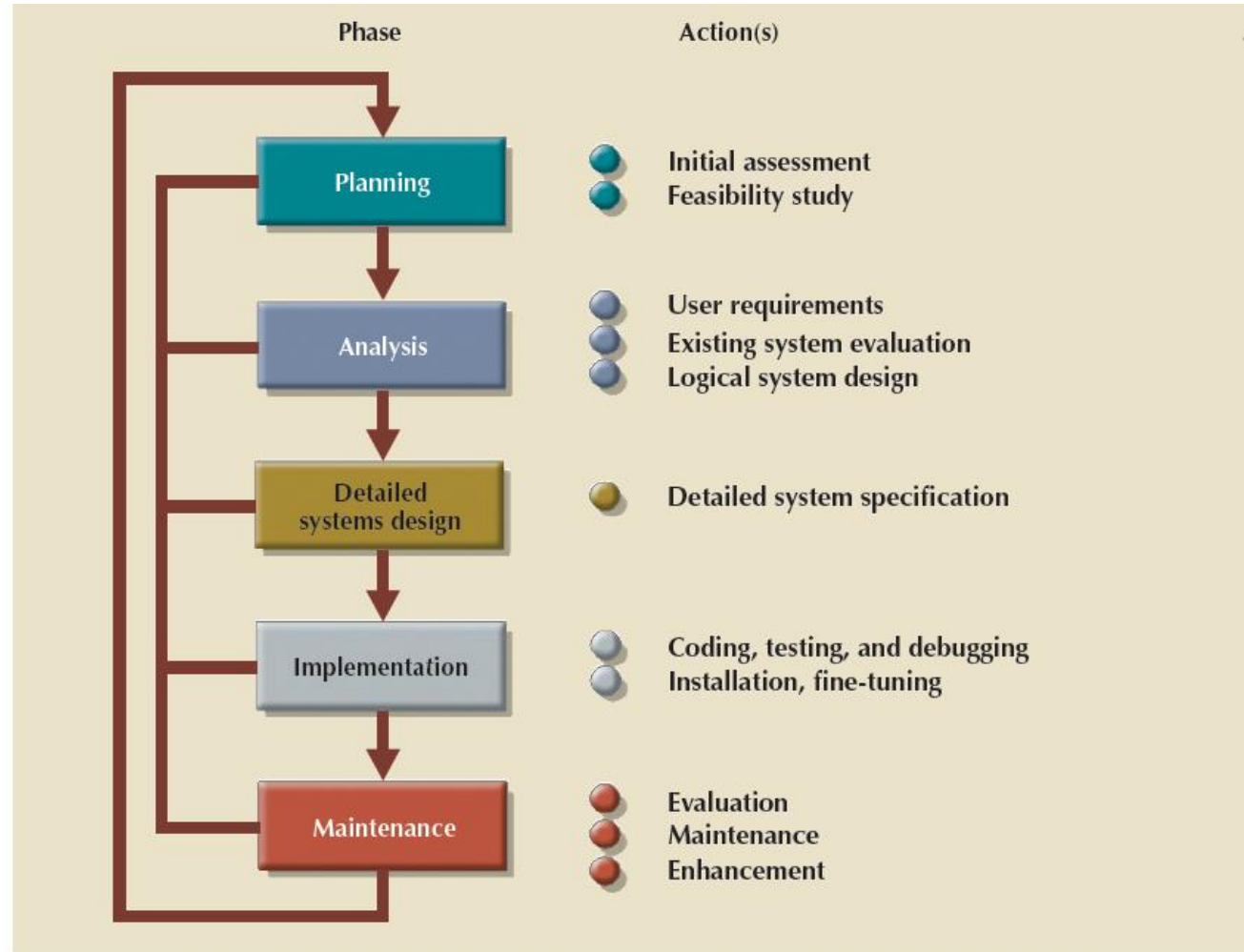
Building an Information System



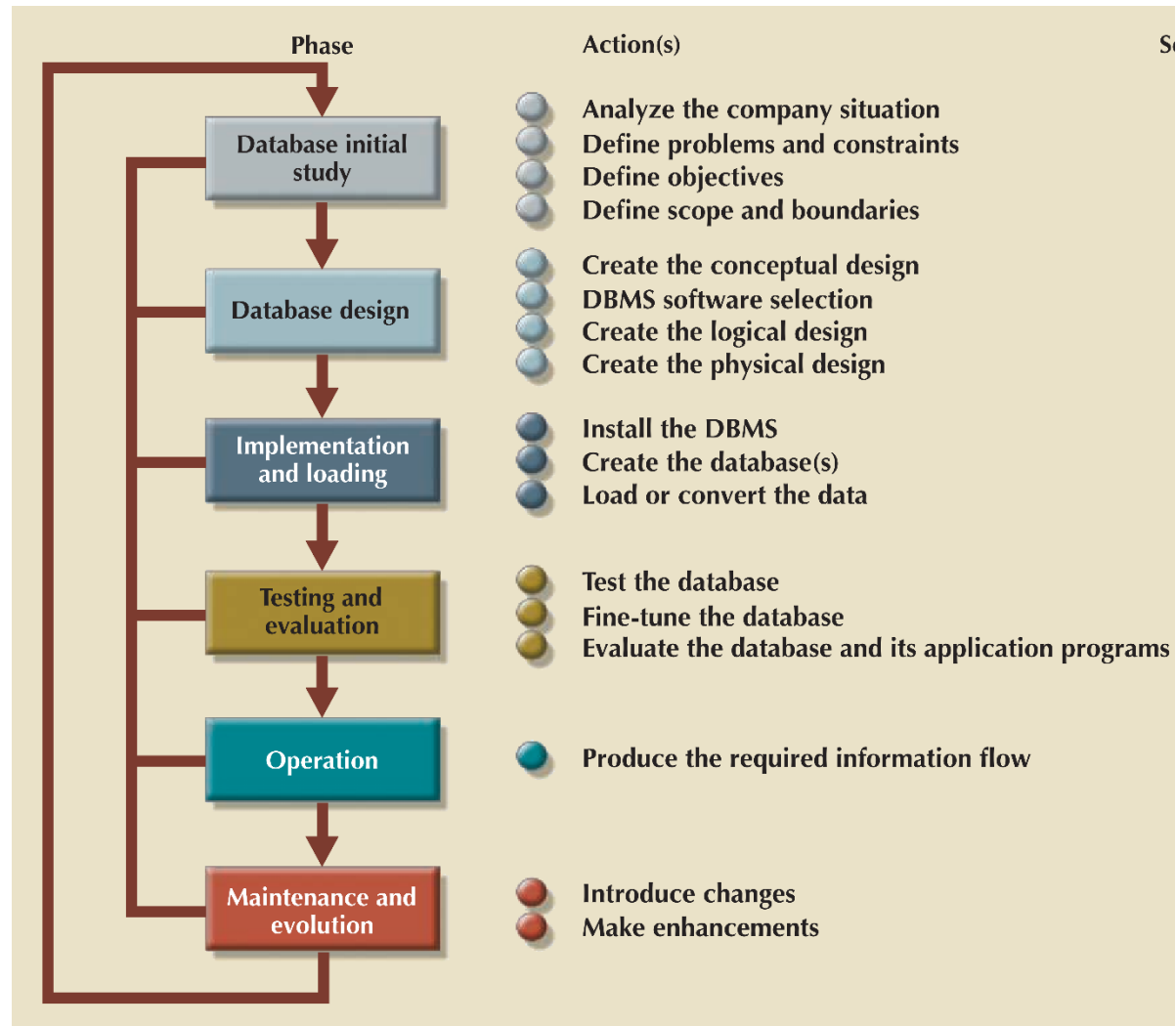
What do we need to
build an information
system?

- 
- ☐ Identify the content of IS
 - ☐ Design IS and database
 - ☐ Store database
 - ☐ Building & Exploiting IS

Systems Development Life Cycle (SDLC)



The Database Life Cycle (DBLC)



☐ Mandatory courses

STT	Mã môn học	Tên môn học
1	CSC10006	Cơ sở dữ liệu
2	CSC12001	An toàn và bảo mật dữ liệu trong HTTT
3	CSC12002	Cơ sở dữ liệu nâng cao
4	CSC12003	Hệ quản trị cơ sở dữ liệu
5	CSC12004	Phân tích thiết kế HTTT
6	CSC12005	Phát triển ứng dụng HTTT hiện đại

Optional courses

STT	Mã môn học	Tên môn học
1	CSC12109	Hệ thống thông tin doanh nghiệp
2	CSC12107	HTTT phục vụ trí tuệ kinh doanh
3	CSC12102	Chuyên đề chọn lọc trong HTTT
4	CSC12105	Thương mại điện tử
5	CSC12106	Tương tác người máy
6	CSC12108	Ứng dụng phân tán
7	CSC10108	Trực quan hóa dữ liệu
8	CSC12103	Chuyên đề Hệ QTCSDL NC
9	CSC12110	Phân tích dữ liệu ứng dụng
10	CSC12111	Quản trị CSDL hiện đại

□ Chuỗi phát triển kỹ năng

Phân tích hệ thống

Thiết kế hệ thống

Cài đặt và
Vận hành hệ thống

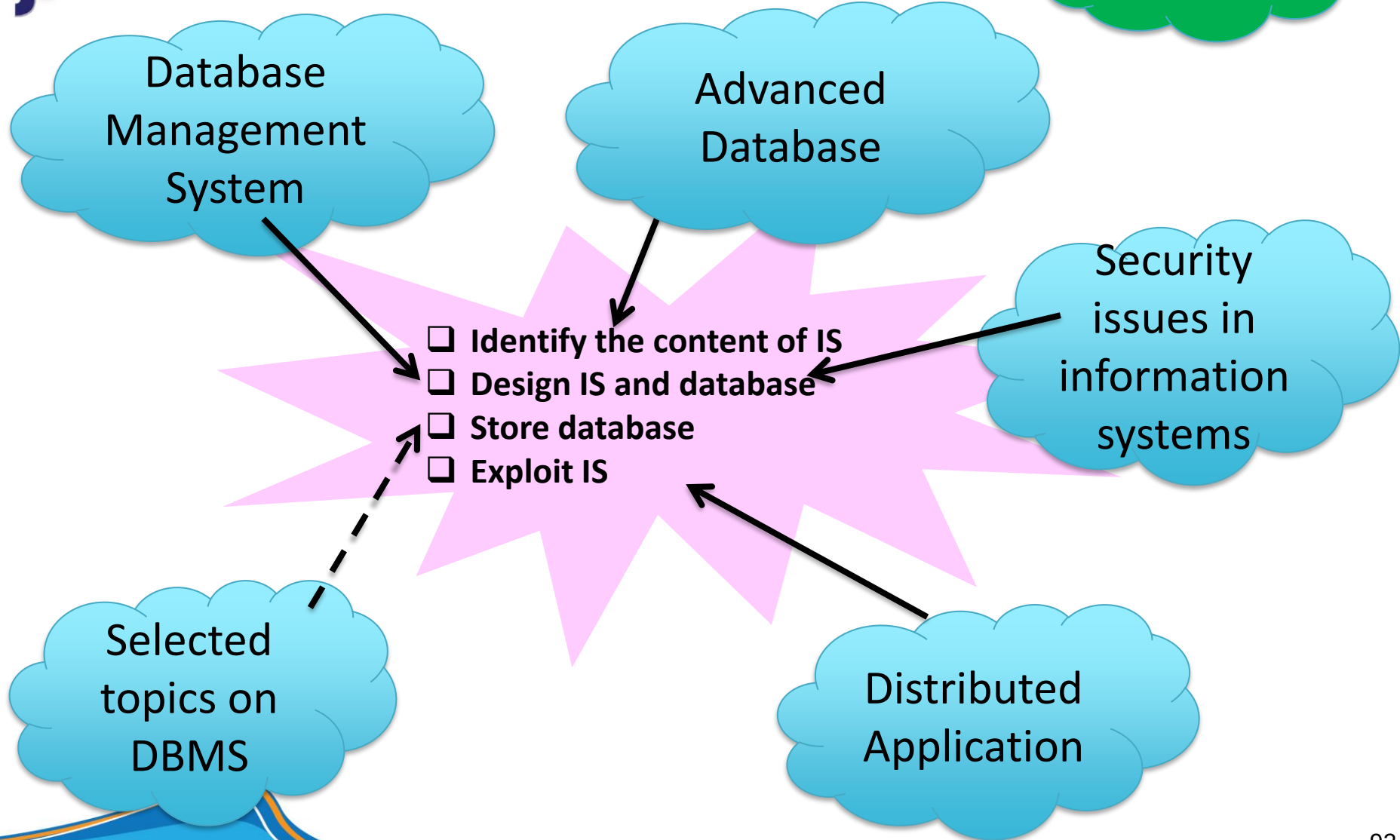


- Hệ QT CSDL
 - MS SQL Server
 - Oracle
 - My SQL

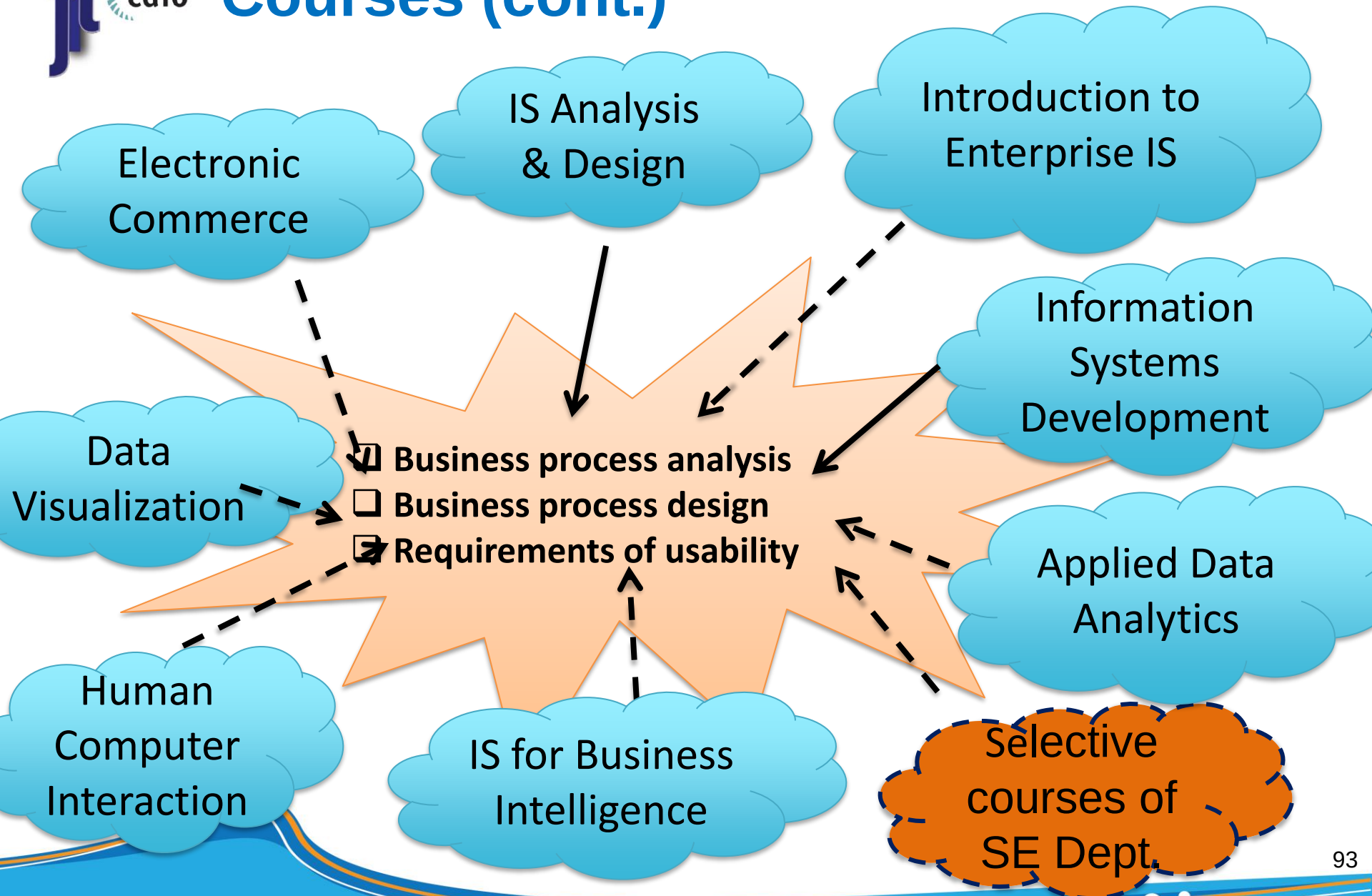
- Ngôn ngữ lập trình
 - C++
 - C#
 - Java

- Công nghệ Web
 - PHP
 - ASP.Net
 - Apache

- Công cụ khác
 - PHP
 - ASP.Net
 - Apache
 - Tool IR
 - Tool Machine Learning



Courses (cont.)



- Information retrieval: Text Mining, Indexing Models, Natural Language Processing for IR, Cross Language IR, Question & Answers Systems, Medical Information Retrieval - (Led by prof. Ho Bao Quoc and Dr. Le Nguyen Hoai Nam, Dr. Nguyen Truong Son)
- Mobile IS: Cooperative Caching Method, Location Management, Mobile Databases, Mobile P2P Applications, Internet of Things –(Led by Dr. Nguyen Tran Minh Thu)
- TEL systems, smart services, knowledge representation, semantic web (Led by Dr. Pham Nguyen Cuong, Dr. Thai Le Vinh)
- System integration, BI, Data analysis, Recommender Systems (Dr. Le Nguyen Hoai Nam, ThS. Ho Thi Hoang Vy)
- Information and data security: Data Confidentiality, Access Control Mechanisms, User Privacy, Data Privacy – (Led by Dr. Pham Thi Bach Huệ)
- Data mining: Data mining in biomedicine – (Led by Dr. Lê Thi Nhàn)

THÔNG TIN GIẢNG VIÊN

TS . PHẠM NGUYỄN CƯỜNG (Trưởng BM)



TIẾN SĨ

TS. Nguyễn Trần Minh Thư
(P.Trưởng BM)

PGS.TS. Hồ Bảo Quốc

TS. Phạm Thị Bạch Huệ

TS. Lê Thị Nhàn

TS. Nguyễn Trường Sơn

TS. Thái Lê Vinh

TS. Lê Nguyễn Hoài Nam

ThS. NCS Vũ Thị Mỹ Hằng

ThS. NCS Hoàng Anh Tú

THẠC SĨ



ThS. Lương Vĩ Minh

ThS. Tuấn Nguyễn Hoài Đức

ThS. Trần Quang Vinh

ThS. Hồ Thị Hoàng Vy

ThS. NCS Hồ Lê Thị Kim Nhung

ThS. Tiết Gia Hồng

ThS. Nguyễn Thị Như Anh

ThS. Phạm Minh Tú

18 Giảng viên

8 Tiến sĩ, 4 NCS (nước ngoài),

6 Thạc sĩ

